K-THEORY OF C*-ALGEBRAS IN SOLID STATE PHYSICS

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1)- C*algebras : an appealing but incomplete tool ?

The subject I would like to present here today pretends to use C*-algebras in concrete problems of modern physics, a mathematical tool which have been liable to controversies among the mathematical physicists for the last twenty years. When D. Testard, R. Lima and I started our program about solving some mathematical problems in the study of disordered systems, five years ago, we naively believed that using C*-algebras was just like using any other tool of mathematical physics like functional analysis or probability theory. However when we began to explain to the experts what we were able to do, not very much I confess, we realized that before trying to convince them, it was necessary to get really hard results. Since we knew perfectly well the limitations of these technics, we understood quite soon that what could be solved with it, was not as spectacular as was implicitly demanded by both the believers and the sceptics in the community. We spontaneously adopted what can be qualified as a schizophrenic attitude : on the one hand we presented an external work as reliable as possible, and on the other hand we kept unpublished for a very long time our results always present in the back of our mind as a guide in performing our program.

What I will explain today have been presented almost in essence at an informal meeting in Marseille during the spring of 1981, apart from the chapter on the quantum Hall effect. Actually some traces of this work can be found in the recent literature. Several results were partially quoted in seminars, conferences [14], reviews [94], or even in some papers [17,56]. We finally published the first part very recently [19]. Reading this last paper will explain the way it has been received by the scientific community. At first sight, the scheme seems very simple and so efficient that it looks miraculous. And when miracles appear the world is immediately divided into two irreducible components : those who are true believers and those who just reject the claims. Fortunately, or perhaps unfortunately depending upon the side one prefers to choose, there is no miracle, at least in physics. In our problem, even though the scheme looks very simple, when one starts the proofs, one realizes immediately how heavy they are : all the machinery of C*-algebras and algebraic geometry is needed, and one looses quite rapidly the original physical intuition! Moreover many technical details remain unsolved or require very complicated and winding proofs. Clearly C*-algebras are very appealing mathematical tools but they must be used together with other technics to become efficient.

One can say that C*-algebras started their career in mathematical physics in the early sixties with the Haag-Kastler axioms [48] of what is known today as the Structural Field Theory. The hope was that the miraculous aspect of these objects would solve the mysterious difficulties encountered in Relativistic Quantum Field Theory. Twenty years later it is quite clear that the flu went the other way : more have been learned from the physics to understand the mathematical tool than the converse. Moreover, very few came from the original motivation namely the understanding of the particle physics. On the contrary, like for many advances in physics during the last thirty years, the concrete understanding came from fields connected with Condensed Matter Physics. For example, the KMS condition, one of the major step in the Tomita-Takesaki theory [M29] and in the classification of type III factors by A. Connes [25], was introduced in the field by R. Haag, H. Hugenholtz & M. Winnink [47]. This improvement came after several mathematical physicists who started there work with the axiomatic Quantum Field Theory, had changed their mind and decided to study the Statistical Mechanics [M24]. The attempt of J. Glimm and A., Jaffe [45] in using C*-algebras in constructive quantum field theory was soon abandoned and replaced by the probabilistic approach[M10,M27].

However eight years ago a breakthrough was performed by A. Connes [26,27,30] who associated to any foliated manifold a canonical C*-algebra, playing the role of the basic object needed to apply the methods of algebraic geometry and to get topological invariants as well as a cohomology theory. He contributed in giving a concrete content to the K-theory, and created the cyclic cohomology, to take into account this new situation [28-34]. He also emphasized that non commutativity of the algebras was closely related to the ergodicity of the foliation, or equivalently to the fact that the set of leaves was not endowed with a non trivial topology.

At about the same time, algebraic geometry entered in quantum field theory with the contribution of mathematicians to the classification of instantons in gauge field theories [M15]. Later on, they also interpreted the anomalies (as Adler's one) in term of topological obstructions. More recently the supersymmetries offered the mathematicians a formal tool for calculating simply the formulæ appearing in the Index theory. Nevertheless the previous works are all performed on the ground of a semi classical quantization by mean of hypothetical Feynman path integrals: the algebraic geometry concerns usual fibre bundles, on usual manifolds. Nothing in this approach is essentially non commutative. The reason is that nobody knows how to built a theory for describing gauge fields in four dimensions, and one has the hope that something of the topological obstructions will remain in the final theory if any. However literally speaking there is an apparent contradiction between the fact that topological obstructions require smooth configurations of gauge fields, whereas it is well known in constructive field theory that almost surely no configuration is smooth.

Topological invariants actually reflect the properties of the observable algebra, more than those of the configurations. The Quantum Hall Effect (QHE) may be a good laboratory to test such an idea. What will be presented here concerns only the ordinary QHE, namely a one particle theory. The quantum mechanics of such a system is well understood. Moreover its has been demonstrated that, when the Fermi energy belongs to a gap, and at zero temperature, the Hall conductivity can be interpreted (in e^2/h units) as the Chern character of some fibre bundle above the Brillouin zone, at least when the magnetic flux through a unit cell, correctly normalized, is rational [99,9]. When it is irrational, the C^*-algebra of observables becomes non commutative in a non trivial way : it is type II as was remarked a long time ago by A. Grossmann [46]. It is no longer possible to speak about Brillouin zone, and the previous interpretation of the Hall conductivity becomes meaningless. Nevertheless, thanks to the A. Connes proposal, the C^*-algebra of observable can be interpreted as the algebra of continuous functions on an hypothetical Brillouin zone which does not exist as a geometrical object but has sufficiently many continuous "non-commutative" functions to be studied by the technics of the algebraic geometry. In particular, the Chern character still exists and varies continuously with the magnetic field, in such a way that the semiclassical approach gives the correct answer.

It is widely admitted that the fractional QHE comes from many bodies interactions. A correct mathematical theory of it should require a Fock space, and a second quantized algebra of observable. It is therefore a (non relativistic) quantum field theory. Nevertheless there is a structural question here related to the calculation of the Hall conductivity : somewhere there must be some topological invariant. Is this related to the differential algebra introduced by Arveson [4] and used in the definition of the cyclic cohomology [32,33]? Is the vacuum a closed current in the sense of Connes ? Is the Hall conductivity given by a Chern character in the many body problem ?

Besides the internal interest of the physical problem, which goes far beyond the analysis I will give here, it seems that investigating easier situations which are better understood in the physical set up than the intricate questions in particle physics, is an efficient way of improving our understanding of the mathematical tool, and gives the hope that in the next future, paradoxes will be solved. For this reason I feel that mathematical physicists should take more seriously the mathematical study of simple , realistic models of condensed matter physics where topological invariants exist in order to understand much better what to do in more ambitious questions like the structure of gauge field theories.

Acknowledgments : **This work benefited from many discussions both encouraging or discouraging and from several helps to visit many Institutes, giving me the opportunity to meet many experts. The list of contributors to these exchanges would be too long to be reproduced here, and they have been quoted in previous works. Let me thank them collectively here. Let me however express my gratitude to A. Connes whose constant enthusiasm, efficiency and patience in explaining the properties and the importance of C* algebras, convinced me to start this program and to go beyond the formalism it offered at the beginning.**

2)- Why are C*-algebras needed to describe disordered systems ?

A real sample used in experimental solid state physics is made of a finite assembly of atoms bound together by electrostatic interactions. However, even though finite and relatively small (few millimeters in size, sometime smaller in crystallography), any sample contains so many atoms that it is better described via the use of the infinite volume limit. On the other hand, it appears homogeneous at large scale while at atomic scale the disorder breaks any translation invariance.

When studying electronic properties of such a crystal, it is commonly admitted that a one electron approximation is usually quite good. Collective properties of the electrons gas are described through the model of a perfect Fermi gas. The disorder appears simply as external forces coming from the random positions of the atoms or impurities. In this set up, the Schrödinger operator for an electron, acting on the space $L^2(\mathbb{R}^D)$ (D being the dimension of the crystal), is given by an effective hamiltonian of the form :

$$(1) \qquad H = -(\hbar^2/2m)\Delta + \Sigma_i \; v_i(x-x_i)$$

where v_i is the potential created by the i^{th} atom or impurity and x_i denotes its position.

The disorder may have several sources. One is given by the randomness of the position of the atoms. This randomness may come from the occurrence of many defects due to the way the sample has been prepared. It may as well come from structural reasons, namely the thermodynamical equilibrium favors the occurrence of some disorder : this is the case for amorphous materials (glasses) or quasi-crystals. Another source of disorder is also given by the impurities which modify the atomic potential at random positions.

In any case, to describe such a system in full generality, one usually introduces a probability space (Ω, Σ, μ) , the points of which labelling the hamiltonian and describing in an implicit way the configuration of the material. In other words H becomes a function of $\omega \in \Omega$. For obvious mathematical reasons, one demands that H be a measurable function of ω, at least in the strong resolvent sense (we recall that Borel sets are the same for the norm and the strong topology in the algebra of bounded operator in a separable Hilbert space). To describe the macroscopic homogeneity of the material we just remark that translating the electron in the sample is equivalent to translating the atoms backward and since the sample looks almost the same at any place, this is just changing the configuration ω in Ω. Therefore, there must be an action $\omega \rightarrow T_x\omega$ of the translation group $\mathbb{R}^D$ on Ω such that, if U(x) is the unitary operator representing the translation by x in the Hilbert space $L^2(\mathbb{R}^D)$ then :

$$(2) \qquad U(x) . H_\omega . U(x)^* = H_{T_x\omega} \qquad \text{(covariance)}$$

This action will be at least measurable and will satisfy the group property $T_xT_y = T_{(x+y)}$. As we shall see in the end of this section,we may assume Ω to be actually a compact space and the $\mathbb{R}^D$ action to be given by a group of homeomorphisms. The probability measure will serve later on in dealing with "self averaging" quantities.

The framework can be applied to the study of the phonon spectrum as well. For indeed, phonons are elementary mechanical excitations of the atoms around their equilibrium position. In the approximation where there is no phonon-phonon interaction, they are described quantum mechanically as a free Bose field, and only the one phonon hamiltonian must be considered : we must solve the corresponding classical problem. The classical eigenmodes are just eigenvalues of a discrete Schrödinger-like operator with random coefficients. Again the randomness comes from the disorder inside the sample. In any case this operator will obey a covariance condition given by (2).

Even though in practice a sample is given from the beginning and will not change at low temperature during the experiment, the physical energy operator is not the operator H_ω alone corresponding to the given configuration of the disorder. Otherwise it would imply the choice of an origin of coordinates inside the sample, preventing the use of its homogeneity : there is no physical reason "a priori" to prefer one origin instead of one another ! Therefore the observable algebra must contain not only H_ω but also the whole family of its translated. Since in general two elements of this family do not commute **the observable algebra will be non commutative in a fundamental way**.

When performing calculations in quantum mechanics, we need to compute several operators starting from the basic observables. In particular, we will need operators defined through series expansion and the question of convergence will be addressed. Therefore we need to define on the set of observables an algebraic structure which will allow us to make computations, and also a topology which will be essential in proving convergence of infinite series. Clearly the algebra obtained will be a *-algebra since we need to distinguish real numbers from the complex ones. However there is a wide choice of possible topologies depending upon the technical point of view we will choose. Nevertheless there is a canonical choice namely **a topology which is of purely algebraic origin**. There are two kinds of such topological *-algebras : the C*-algebras and the W*-algebras. For in the former case the norm of an element A is nothing but the spectral radius (a purely algebraic object) of A^*A. On the other hand a W*-algebra is a C*-algebra with a predual (which is unique) [M25]. Equivalently it is a weakly closed *-subalgebra of $B(\mathcal{H})$. The von Neumann theorem [M6], shows that it coincides with its bicommutant (again an algebraic property even though it may depend upon the representation). As we shall see the W*-algebra built out of the translated of the original hamiltonian is too large to contain any relevant informations. For this reason we shall prefer the C*-algebra generated by the hamiltonian and its translated. In a sense this algebra is the smallest object which contains all the relevant physical informations.

The next step in describing the formal framework consists in answering the question whether it is possible to recover the nature of the probability space Ω from the physical datas. Since the translated of the hamiltonian must belong to the algebra of observables, it is natural to built Ω out of the family of self adjoint operators we obtain in this way. For this reason we introduce the following mathematical criterion to describe what we call "homogeneity".

<u>Definition</u> : A bounded operator A on $L^2(\mathbf{R}^D)$ is called homogeneous if the family of its translated $\{U(x)AU(x)^* \,;\, x \in \mathbf{R}^D\}$ has a compact strong closure.

◇

Strong compactness means that for any finite family of vectors $\varphi_1,\dots,\varphi_N$, and any $\varepsilon > 0$, there is a finite set $x_1,\dots,x_M$ in $\mathbf{R}^D$, such that each vector $U(x)\,A\,U(x)^*\varphi_i$ $(x \in \mathbf{R}^D)$ stays within the distance ε of one of the vectors $U(x_j)\,A\,U(x_j)^*\varphi_i$ $(1 \le j \le M)$ in $L^2(\mathbf{R}^D)$. In other words, after translation, the operator A reproduces itself everywhere up to ε. This is why we called it homogeneous.

Actually this notion of homogeneity is quite weak. Indeed the operator of multiplication by a continuous function vanishing at infinity is homogeneous according to our definition (it corresponds to the potential created by finitely many impurities). Another example is given by the following :

<u>Proposition 1</u> : if $V \in L^\infty(\mathbf{R}^D)$ then the resolvent of the Schrödinger operator $H = -\Delta + V$ is homogeneous. ◇

To prove this result, we use a Neuman series expansion and the expression of the Green function of the free laplacian to show that the mapping $V \to (z-H)^{-1}$ is strongly continuous if we endow $L^{\infty}(\mathbf{R}^D)$ with the weak topology. Since any ball in $L^{\infty}(\mathbf{R}^D)$ is weakly compact and since translating the resolvent is equivalent to translating the potential alone, the set of translated of the resolvent lies in a strongly compact set.

The main interest of this definition lies in the following construction. The hull $\Omega = \mathrm{Hull}(A)$ of a homogeneous operator A is the strong closure of the set of its translated. By hypothesis it is a compact space. Moreover it is translation invariant. Thus $\mathbf{R}^D$ acts on Ω through a group of homeomorphisms which will be denoted by $\{T_x\,;\, x\in \mathbf{R}^D\}$: any $\omega \in \Omega$ is a bounded operator as the strong limit of some sequence $A_k = U(x_k)AU(x_k)^*$ and we have $T_x\omega = U(x)\omega U(x)^*$. Therefore, the map $(x,\omega) \to T_x\omega$ is continuous. In this way, associated to any homogeneous operator, we get a dynamical system which will also be called the hull of A.

From the proof of proposition 1 it follows that if V belongs to $L^{\infty}(\mathbf{R}^D)$ the hull of the resolvent of $H = -\Delta + V$ is homeomorphic to the weak closure of the set of translated of V in $L^{\infty}(\mathbf{R}^D)$. For instance if V is continuous and vanishes at infinity, the hull is homeomorphic to the one point compactification of $\mathbf{R}^D$, namely a D-sphere, and $\{\infty\}$ is the set of non wandering points of the flow generated by the translations.

More interesting is the case for which V is almost periodic in the sense of Bohr [M4]. Generalizing this case leads to the following definition :

<u>Definition</u> : A bounded operator A on $L^2(\mathbf{R}^D)$ is called almost periodic if the family of its translated $\{U(x)AU(x)^*\,;\, x\in \mathbf{R}^D\}$ has a compact norm closure.

◇

By mimicking the construction of the hull of an almost periodic function on $\mathbf{R}$, one can show that the hull of an almost periodic operator is an abelian compact group, and $\mathbf{R}^D$ acts as $\omega \to \omega + \gamma(x)$ where γ is a continuous homomorphism from $\mathbf{R}^D$ into Ω with a dense image. Following Mackey's notion of virtual group [71,72], such a dynamical system will be called an "abelian virtual group".

Let us note that the previous construction is universal because any kind of dynamical system can be obtained in this way. For indeed if Ω is a compact space endowed with an $\mathbf{R}^D$ action through a continuous group T of homeomorphisms, and if v is any continuous function on Ω, then for each ω_0 in Ω, the function $x \to v(T_{-x}\omega_0) = V(x)$ is uniformly continuous and the hull of the Schrödinger operator it defines is homeomorphic the dynamical system (Ω,T).

In our purpose, for most interesting examples, taking the W*algebra generated by the family of translated of a homogeneous operator A in $L^2(\mathbf{R}^D)$ will give rise to the full set of bounded operators. Therefore no specific property of the physical system we want to investigate will be seen out of it. For this reason we shall prefer to take the smallest possible *algebra which contains A and all its translated, and which is closed in the norm topology : let $C^*(A)$ be this C^*-algebra.

Using a "3ε argument", one can show, that if A is homogeneous, so does any element of $C^*(A)$, for the product is strongly continuous on bounded sets. Moreover, if $\omega \in \Omega$, and if $(x_k)_{k>0}$ is a sequence in $\mathbf{R}^D$ converging to ω in Ω (we consider $\mathbf{R}^D$ as a subset of ω here), then for any $B \in C^*(A)$ the sequence $U(x_k)\, B\, U(x_k)^*$ converges strongly to some operator B_ω which depends only upon ω and B. The map π_ω: $B \in C^*(A) \to B_\omega$ is a *representation of $C^*(A)$ into the space of bounded operators on $L^2(\mathbf{R}^D)$. Moreover, the map π: $\omega \in \Omega \to \pi_\omega$ is pointwise strongly continuous and it satisfies the following covariance condition :

$$(2) \qquad U(x)\, \pi_\omega(B)\, U(x)^* = \pi_{T_x\omega}(B) \qquad \omega \in \Omega,\ x \in \mathbf{R}^D,\ \ B \in C^*(A)$$

As a matter of fact, it cannot be more than strongly continuous in general for we have :

<u>Proposition 2</u> : the representation map π is pointwise norm continuous if and only if A is almost periodic, or equivalently if its Hull (Ω,T) is an abelian virtual group.

◇

The proof is just an elementary application of the definition. We now address the question whether one can compute explicitly the algebra $C^*(A)$. To each dynamical system (Ω,T) is associated a natural C*algebra, namely the cross-product of $C(\Omega)$, the algebra of continuous functions on Ω, by $\mathbf{R}^D$ through T [M23]. What is the connection between $C^*(A)$ and $C^*(\Omega,T)$?

Let us recall the construction of $C^*(\Omega,T)$. The vector space $C_c(\Omega \times \mathbf{R}^D)$ of continuous functions on $\Omega \times \mathbf{R}^D$ with compact support is endowed with a structure of *-algebra as follows :

$$(3) \qquad a.b(\omega,x) = \int d^Dy\ \ a(\omega,y)\, b(T_{-y}\omega,x-y) \qquad a^*(\omega,x) = a(T_{-x}\omega,-x)^*$$

We consider now on $L^2(\mathbf{R}^D)$ the family of *representation π_ω defined as :

$$(4) \qquad \pi_\omega(a)\, \psi(x) = \int d^Dy\ \ a(T_{-x}\omega,y-x)\, \psi(y)$$

$\pi_\omega(a)$ is a bounded operator. A C*norm is then given by :

(5) $$\| a \| = \sup_{\omega\in\Omega} \| \pi_\omega(a) \|$$

Then $C^*(\Omega,T)$ is nothing but the completion of $C_c(\Omega\times\mathbb{R}^D)$ under this norm. If $a \in C^*(\Omega,T)$, $\pi_\omega(a)$ is a strongly continuous function of ω and satisfies the covariance condition (2). Therefore, since Ω is compact, it is a homogeneous operator. We will say that A is affiliated to $C^*(\Omega,T)$ if there is ω_0 in Ω, and $a \in C^*(\Omega,T)$ such that $A = \pi_{\omega_0}(a)$.

Not every homogeneous operator A is affiliated to the C*algebra of its hull. Two conditions are required for this: first of all A must be "diagonal dominant" namely it must be the norm limit of a sequence A_n of operator for which there is $R_n > 0$, such that $\langle\psi| A_n \varphi\rangle = 0$ whenever the distance between the supports of ψ and φ is greater than R_n. Secondly A must be regular, namely it is the norm limit of operators with a continuous kernel. For instance the operator of multiplication by V is never affiliated with C*(Hull(V)) even if it is smooth. However we get :

<u>Proposition 3</u>: If $V \in L^\infty(\mathbb{R}^D)$ then $H = -\Delta + V$ is affiliated to $C^*(\Omega,T)$ where (Ω,T) is the Hull of the resolvent of H.

◇

To prove this result it is sufficient to compute the kernel associated to the operator $A(f)=f(-\Delta)Vf(-\Delta)$ where f is some smooth function with an L^1 positive Fourier transform. This operator is strongly continuous with respect to V if $L^\infty(\mathbb{R}^D)$ is endowed with the weak topology. Since the weak closure of the set of translated of V is homeomorphic to the Hull of the resolvent of H this shows that A is actually affiliated to C*(Hull(H)). Now, it is clear also that if we now take for f the Fourier transform of $(1+p^2)^{-1/2}$, A is a norm limit of operators of the form A(g) with g rapidly decreasing. Then it is still in the C*algebra. Taking V=1 we get the resolvent of $-\Delta$ which is therefore affiliated to C*(Hull(H)), and using a Neuman expansion of the resolvent of H we get the result.

3)- The tight binding representation :

In this section we want to take into account that the samples studied previously look like discrete lattices, even if there is some disorder in them. As we shall see, there is a mathematical counterpart, namely the Poincaré section [M26], which transforms a continuous flow into a discrete one. Correspondingly there will be a C*algebra for the discrete flow. The converse operation is called the suspension of a discrete flow (or of the C*algebra).

Let us consider the model given in §2-eq(1). Let us assume for simplicity that there is only one species of atoms. This is not actually a restriction if all species have an homogeneous spatial distribution. We call R the smallest distance between any pair of atoms in the sample. We can think of H as depending on R by setting :

(1) $$V(x) = \Sigma_i\, v(x-Rx_i))$$

where v is the atomic potential. If the disorder is not too big, we can label the atom by $i \in \mathbf{Z}^D$: it is not too faraway from the ideal position it would occupy in a perfect crystal. Now, v is attractive, and therefore the atomic hamiltonian $H_a = -\Delta + v$ must have bound states in order to bind the electrons. Let E be the energy of such a bound state and let ψ be the corresponding wave function. We will also call δE the energy distance between E and the energy level closest to E. If we choose the energy to be zero at infinity, where the atomic potential vanishes, the wave function ψ decreases exponentially fast at infinity, and the rate of decreasing is $|E|$ [M1]. Therefore if R is big enough, the functions $\psi_i(x) = \psi(x-Rx_i)$ are approximate eigenfunctions of $H = -\Delta + V$, and E is the corresponding eigenvalue. Actually this is not quite correct unless we take the limit $R \to \infty$. To take into account the corrections let us consider the matrix L on $\mathbf{Z}^D$ defined by :

(2) $$L_{ij} = \langle \psi_i \mid \psi_j \rangle = \delta_{ij} + O(e^{-ER}) \qquad \text{as } R \to \infty$$

This matrix is non negative, and there is R_0 such that if $R > R_0$, $\|L-\mathbf{1}\| < 1$ and therefore $L^{-1/2}$ is well defined. Now we set :

(3) $$\varphi_i = \Sigma_j\, (L^{-1/2})_{ji}\, \psi_j \qquad \Rightarrow \qquad \langle \varphi_i \mid \varphi_j \rangle = \delta_{ij}$$

Thus we get an orthonormal basis of the subspace $\mathcal{H}$ of $L^2(\mathbf{R}^D)$ spanned by the ψ_i's. Moreover, $\|\varphi_i - \psi_i\| = O(e^{-ER})$ as $R \to \infty$.

Let now P be the projection onto $\mathcal{H}$. Then for R large enough the spectrum of PHP is contained in the open interval J of width $\delta E/2$ centered at E, which does not meet the spectrum of (1-P)H(1-P). To study the spectrum of H which is contained in this interval, there is a well known method, the first step of which goes back to Schur [101]. It was developed by Feschback and is nowadays called the projection method [101]. Let us define the following energy dependent operator (where Q=1-P):

$$(4) \qquad H(z) = PHP + PHQ\,[z-QHQ]^{-1}QHP$$

As a function of z it is holomorphic on the complement of the spectrum of QHQ, and therefore it is holomorphic in a neighbourhood of the interval J. The main result about it is the following:

<u>Proposition 4</u>: (i) An energy ε belongs to the part of the spectrum of H contained in J if and only if ε belongs to the spectrum of $H(\varepsilon)$.

(ii) ε is an eigenvalue of H with eigenvector φ if and only if ε is an eigenvalue of $H(\varepsilon)$ with eigenvector $P\varphi$. Then φ can be recovered from the formula:

$$(5) \qquad Q\varphi = [\varepsilon-QHQ]^{-1}QHP\varphi$$

◇

From the definition of ψ_i it is simple to show that for any Φ in $L^2(\mathbb{R}^D)$:

$$(6) \qquad \langle\Phi|H\psi_i\rangle = E\,\langle\Phi|\psi_i\rangle + \sum_{j\neq i} \langle\Phi|v(\,.-Rx_j)\psi_i\rangle = E\,\langle\Phi|\psi_i\rangle + O(\|\Phi\|\,e^{-ER})$$

In particular it follows from (2),(3) & (6) that QHP is a bounded operator and that the correction $PHQ\,[z-QHQ]^{-1}QHP$ is bounded by $O(e^{-2ER})$ as $R\to\infty$ uniformly for $z\in J$. Now, the matrix of H(z) in the basis $(\varphi_i)_i$ is diagonal dominant namely:

$$(7) \qquad \langle\varphi_i\,|\,H(z)\varphi_j\rangle = O(e^{-RE|x_i-x_j|}) \qquad \text{as } R\to\infty$$

Let us remark that L_{ij} is actually a function of x_i-x_j, and therefore, there is a function φ in $L^2(\mathbb{R}^D)$ such that $\varphi_i = U(x_i)\varphi$.

What we have done here was to replace the unbounded operator H on the continuum $\mathbb{R}^D$ by an infinite matrix acting on the sequences indexed by the lattice of atoms. As far as we are concerned with the local properties of the spectrum, the two problems are equivalent.

The next step is to introduce a knew C*algebra which will contain the previous matrix. Following A. Connes [27], we need the notion of transversal. Indeed let us consider in Ω the set:

$$\Xi = \text{closure of } \{T_{x_i}\omega_0 \,;\, i \in \mathbf{Z}^D\} \quad \text{where } \omega_0 \in \Omega \text{ represents } V.$$

Ξ is a "transversal" in the following sense (which differs slightly from the Connes definition, to take into account the topology of Ω):

Definition : A closed subset Ξ of Ω is called a transversal if for any continuous function f on $\mathbf{R}^D$ with compact support, the map

(8) $$(\omega,f) \to \nu^{\omega}(f) = \sum_{x:T_{-x}\omega\in\Xi} f(x)$$

is continuous, for the topology of uniform convergence on compact sets.

◇

It is not hard to check that if Ξ is a transversal, for any $\omega \in \Omega$ the set $L(\omega) = \{x \in \mathbf{R}^D \,;\, T_{-x}\omega \in \Xi\}$ is actually discrete. In addition, given a bounded set Λ in $\mathbf{R}^D$, if $N(\omega,\Lambda)$ denotes the number of points (or atoms) of $L(\omega)$ in Λ, then by subadditivity, the limit :

(9) $$\rho = \lim_{\Lambda \uparrow \mathbf{R}^D} |\Lambda|^{-1} \sup_{\omega} N(\omega,\Lambda)$$

exists. It is a measure of the density of the atoms in the sample. Now let g be a bounded continuous function decreasing at infinity in such a way that $\|g\| < \infty$ where :

(10) $$\| g \| = \sum_{q\in\mathbf{N}} q^{D-1} \sup_{|x|\geq q} |g(x)|$$

Then one can show that (9) implies $\sup_{\omega} |\nu^{\omega}(g)| \leq \text{const.}\|g\|$, and therefore, since such a g can be approximated in this norm by continuous functions with compact support, $\nu^{\omega}(g)$ is also continuous with respect to ω. Let B be the Banach space of such g's. Now let us choose v in $L^{\infty}(\mathbf{R}^D)$ and such that $\| v \| < \infty$ (where the supremum is replaced by the essential one in (10)). For any continuous function f with compact support, the convolution $f_* v$ belongs to B, which shows that $\langle V_{\omega} | f \rangle = \langle \nu^{\omega}(v) | f \rangle = \nu^{\omega}(f_* v)$ is continuous in ω. Again since $\| v \|$ is finite, V_{ω} is essentially bounded, which shows that it is weakly continuous with respect to ω.

As we already quoted, the flow on Ω can be restricted to Ξ by mean of the "first return map" introduced by H. Poincaré [M21] in the 19th century to study the existence and the stability of periodic orbits of the planets. In our many dimensional framework, the first return map does not exist, but is replaced by the notion of groupoid of the transversal Ξ [27]. Let $\Gamma(\Xi)$ be the (closed) subset of $\Omega \times \mathbf{R}^D$ made of the elements $\gamma = (\omega,x)$ such that $\omega \in \Xi$ and

$T_{-x}\omega \in \Xi$. It is a locally compact groupoid in the sense that $\gamma=(\omega,x)$ and $\gamma'=(\omega',x')$ can be multiplied provided $\omega'=T_{-x}\omega$; in this case we get $\gamma\circ\gamma'=(\omega,x+x')$. A *algebra can be constructed through continuous functions on $\Gamma(\Xi)$ with compact support with the following rules :

$$(11) \quad a.b(\omega,x)=\sum_{y\in L(\omega)} a(\omega,y)b(T_{-y}\omega,x-y) \qquad a^*(\omega,x)=a(T_{-x}\omega,-x)^*$$

As in the previous section, let $C^*(\Xi)$ be the completion of this algebra with respect to the C^* norm $\|a\|=\sup_{\omega\in\Xi}\|\eta_\omega(a)\|$ where η_ω is the *representation on the space $l^2(L(\omega))$ of square summable sequences indexed by $L(\omega)$ and defined by :

$$(12) \qquad \eta_\omega(a)\,\psi(x)=\sum_{y\in L(\omega)} a(T_{-x}\omega,y-x)\,\psi(y)$$

These representations satisfy also a covariance condition of a different kind. Namely, for $\gamma=(\omega,x)$ in $\Gamma(\Xi)$, let $U(\gamma)$ be the unitary operator from $l^2(L(T_{-x}\omega))$ into $l^2(L(\omega))$ given by $U(\gamma)\,\psi(y)=\psi(y-x)$. Then we get :

$$(13) \qquad U(\gamma)\,\eta_{T_{-x}\omega}(a) = \eta_\omega(a)U(\gamma)$$

The main application of this formalism is given by :

Proposition 5 : Let $h(z)$ be the function on $\Gamma(\Xi)$ defined from (7) by :

$$(14) \quad h(z;\omega,x)=\langle \varphi \mid H_\omega(z)\,U(x)\varphi \rangle$$

Then $h(z)$ belongs to $C^*(\Xi)$ for any z for which $H(z)$ is defined. Moreover the matrix of $H_\omega(z)$ in the basis $\{\varphi_i\}$ and of $\eta_\omega(h(z))$ in the basis $\{x_i\}$ coincide.

◇

We will not give the proof of this fact, for (14) is just another way of writing $H_\omega(z)$.

To finish this section let us give one example of such reduction, which is sometime called a "lattice approximation" or also in Solid State Physics a "tight binding representation" because when (4) is valid, the electrons are essentially localized near the bottoms of the potential and are tightly bound to the atoms. Our first example will be a quasi periodic Schrödinger operator in one dimension [see in particular 16]; let v be a smooth function on $\mathbb{R}$ decreasing rapidly to zero at infinity. The potential is given by :

$$(15) \qquad V_\xi(x) = \sum_{n\in\mathbf{Z}} v(x-x_n) \qquad x_n = n + g(\xi)-g(\xi-n\alpha)$$

where g is a continuous function of one variable ξ periodic of period one and α is an irrational number. Then g can be seen as a continuous function on the one dimensional torus $\mathbf{T} = \mathbf{R}/\mathbf{Z}$. To built the hull we define on $\mathbf{T}\mathrm{x}\mathbf{R}$ the map $F(\xi,t) = (\xi+\alpha,t+f(\xi))$, where $f(\xi) = 1+g(\xi+\alpha)-g(\xi)$. Then the function $\mathbf{V}(\xi,x) = V_\xi(x)$ given by (15) is F invariant and continuous on $\mathbf{T}\mathrm{x}\mathbf{R}$. Therefore it defines a continuous function on the quotient space $\Omega=\mathbf{T}\mathrm{x}\mathbf{R}/F$. A flow on Ω is defined by taking the quotient of the translation $T_y:(\xi,x)\to(\xi,x+y)$ (which commutes to F), and we get $V_\xi(x-y) = \mathbf{V}(T_{-y}(\xi,x))$. This shows that Ω is actually the hull. The set $\mathbf{T} = \{(\xi,x) \in \Omega\ ;\ x=0\}$ is a transversal, and the first return map is nothing but the rotation R_α by α. Then it is easy to check that the C* algebra of this transversal $C^*(\mathbf{T}, R_\alpha)$ is generated by two unitaries U and V such that $UV = e^{i\,2\pi\alpha}\,VU$ as was firstly proposed by M. Rieffel [86]. The reduced hamiltonian H(z) has a matrix indexed by $\mathbf{Z}$ which can be seen as an element of $C^*(\mathbf{T},R_\alpha)$. A special case is provided if we can approximate H(z) by a tridiagonal matrix, and if in addition only the first Fourier coefficients are big, one gets :

$$(16) \qquad h(z) = a\,(U + U^*) + b(V + V^*) \qquad a,b \in \mathbf{R}$$

This operator is called the Almost Mathieu operator and has many interesting properties [M16,M28,16,18,23,41,53,94], as we shall se later on.

Our second example is given by the recently discovered quasicrystals [91]. The potential seen by an electron in a quasicrystal can be described in much the same way. However the precise structure of the hull is usually quite complicated : the quasicrystal of AlMn originally discovered had a symmetry of order five. In the original work of D. Schechtman, I. Blech, D. Gratias, J.V. Cahn [91], it was recognized that 6 integers were necessary to label the positions of the peaks of intensity in the X-ray picture. A model has been proposed recently [39,40,65,70] which describes the diffraction pattern observed experimentally. Even though the method leading to its construction was already known before [64], this model have the advantage of being very simple and efficient. Since this crystal exhibits a diffraction pattern having the icosahedral symmetry, and since 6 integers are needed to label the positions of what one believes to be the atoms (there is still a problem in interpreting the nature of the white spots on the pictures), we will look at a 6 dimensional representation of the icosahedral group. This group leaves the 6 dimensional hypercubic lattice $\mathbf{Z}^6$ invariant. Moreover it has two 3-dimensional invariant subspaces. Let Π and Π' be the corresponding eigenprojections. Of course since there is no translation invariant 3-dimensional lattice having a symmetry of order 5, these eigensubspaces are not rationally oriented with respect to the lattice $\mathbf{Z}^6$. Let Δ be the 6-dimensional closed hypercube of unit length, and $\Omega = \Pi'\Delta$. It has been shown that Ω is a polyhedron called a triacontahedron [40] which has axis of order 2, 3 and 5. The model proposed by Duneau & Katz consists in selecting those points $m \in \mathbf{Z}^6$ such that $\Pi'm \in \Omega$, and to project them on the orthogonal subspace, namely $x_m = \Pi m$. Since $\mathrm{Im}(\Pi)$ is 3-dimensional, we will identify it with $\mathbf{R}^3$ and the x_m's with the positions of atoms. By construction the icosahedral group commutes with Π and Π', and therefore the set of atoms is a (non translation invariant) lattice invariant by this group. If we want to take into account a generic translation in $\mathbf{R}^6$ we get the following electronic potential for the quasi crystal :

$$(17) \qquad V_\xi(x) = \Sigma_{m \in \mathbf{Z}^6} v(x - \Pi(\xi+m))\ \chi_\Omega\,(\Pi'(\xi+m))$$

where χ_Ω is the characteristic function of Ω. Using the previous method, the Hull is the 6-dimensional torus $\mathbf{T}^6 = \mathbf{R}^6/\mathbf{Z}^6$. Then $V_\xi(x)$ is given by a discontinuous function on this torus (because of the characteristic function of Ω) as a function of ξ, and by an action of $\mathbf{R}^3$ in the Π direction. The dynamical system that one obtains is an abelian virtual group. The tight binding representation of our hamiltonian will give rise to the C*algebra of the transversal given by the image of the set $\Omega x\{0\}$ in this torus.

In principle the construction is so general that any kind of flow on a compact manifold could be described. The question is to know whether other examples are to be found in experimental physics. The proposal of Sadoc and Mosseri [76], recently studied in detail by D. Nelson [see 78], seems to indicate that indeed other kinds of flows appears in the description of the short range order in liquids and glassy materials.

4)- **The density of states and Shubin's formula** :

The result of the previous analysis is that a disordered system is described by a self adjoint operator which is affiliated to the C*algebra of a dynamical system given by some compact space Ω, the hull, on which there is an action of $\mathbf{R}^D$. As we showed in the previous section, this continuous dynamical system can be replaced by a discrete one, which, in many cases reduces to the action of $\mathbf{Z}^D$ on some compact space Ξ. Before proceeding, let us note that even in physical situations, the groups $\mathbf{R}^D$ or $\mathbf{Z}^D$ can be replaced by a more general locally compact group. For instance in describing the local structure of glassy systems, Kleman & Donnadieu [61] found that a description through hyperbolic geometry is relevant. In this case we have a $SL(2,\mathbf{R})$ action or even the action of a Fuchsian group [M8]. On the other hand, theoretical physicists like to use a special lattice called the "Bethe lattice" [67], to describe the thermodynamics of lattice models in the limit where the dimension goes to infinity. It is not difficult to convince oneself that the Bethe lattice can be seen as a sublattice of the Poincaré disc, and is generated from one point by a free group with a finite number of parameters. Therefore it appears also as a piece of hyperbolic geometry. Another example is given by disordered systems in which spin effects are taken into account, or systems having more symmetries or quasi symmetries than just the translation one as it is the case for quasicrystals.

For this reason let us consider a compact space Ω on which acts a locally compact group G. In much the same way it is easy to construct a C* algebra for this system. We will avoid the construction (see [M23]) for it is formally the same as before. Let us simply remark that it has an identity operator if and only if G is discrete. There is however one property of G which is essential in what follows : it must be **amenable**. Amenability means that in G, there is an increasing sequence $(\Lambda_n)_{n>0}$ of open sets of finite Haar measure (like hypercubes in $\mathbf{R}^D$) the union of which covering the group, the frontier of which has a negligible measure compare to their volume in the limit $n\to\infty$. In other words **surface effects are negligible compare to volume effects**. More precisely, if $x \in G$, Δ denotes the symmetric difference, and $|\Lambda|$ the volume of Λ with respect with a left Haar measure one demands :

$$(1) \qquad \lim_{n\to\infty} |\Lambda_n \Delta (x.\Lambda_n)|/|\Lambda_n| = 0$$

Such a family of open sets of finite volume is called a "Følner sequence" [M12]. As a matter of fact this condition precisely eliminates the hyperbolic geometry, for it is known that in the Bethe lattice, surface effects are not negligible compare to volume ones !

Given one self adjoint operator on $L^2(G)$ affiliated to the C*algebra of (Ω,G), one defines its integrated density of states (if it exists) as follows :

$$(2) \qquad \mathbf{N}_\omega(E) = \lim_{n\to\infty} |\Lambda_n|^{-1} \#\{\text{eigenvalues of } H_\omega|_{\Lambda_n} \leq E\}$$

Let us mention that the first derivative $d\mathbf{N}_\omega(E)/dE$ of this quantity is accessible to the experiment. This is done either around the Fermi level by measuring the current-voltage characteristics of a diode in which the sample has been put, or by mean of X-ray scattering, which permits to investigate the band structure at smaller energies.

For this definition to be meaningful, it is obviously necessary that the restriction of H to bounded regions exists and have a discrete spectrum. Actually if H is affiliated to $C^*(\Omega,G)$, this is always true. Indeed if $a \in C^*(\Omega,G)$, and if χ is the operator of multiplication by the characteristic function of some bounded open set Λ on $L^2(G)$, then for any $\omega \in \Omega$, $\chi\pi_\omega(a)\chi$ is compact. Thus, the restriction of the resolvent of H on χ is compact. This is enough to show that the resolvent of the restriction is also compact. However the restriction of the resolvent is not the resolvent of the restriction. The main difficulty comes from the definition of the restriction of H to Λ. If H is a Schrödinger operator, this requires to specify the boundary conditions. In general, if G is not discrete, then H must be unbounded and bounded from below, at least locally, in order that its restriction have only finitely many eigenvalues near $-\infty$. On the other hand, the domain of H must at least contain "sufficiently many" functions supported by Λ : the set $D(\Lambda)$ of functions in the domain of H supported by Λ must be dense in $L^2(\Lambda)$. Moreover it is also necessary that $\bigcup_n D(\Lambda_n)$ be a core for H in order to recover H as the limit of the family of its restrictions in the strong resolvent sense. If this is true, since H is bounded from below locally, its restriction on $D(\Lambda)$ has self adjoint extensions (the Friedrichs one for instance). A boundary condition is just the assignment of one self adjoint extension. The independence with respect to the boundary conditions is related to the fact that boundary effects are negligible compare to volume effects, namely to the amenability of G. Yet, we do not know what are the necessary and sufficient conditions on H to insure the existence of the density of states in general. However, there are many concrete cases where this has been proved (see conditions (A) below).

When the integrated density of states of H exists, it is a G-invariant Borel function of ω. In particular, it is almost surely constant with respect to an ergodic G-invariant probability on Ω. On the other hand, it is obviously increasing with respect to E. It has been proved by F. Delyon & B. Souillard [36] that for Schrödinger operators on a lattice $\mathbf{Z}^D$, this function is continuous. Their argument is again related to the amenability of $\mathbf{Z}^D$, and we suspect that for reasonable hamiltonians, the integrated density of states is indeed continuous with respect to the energy. Let us note however that there are C* algebras having projections the kernel of which being continuous with compact support (see the appendix) and therefore it is possible to exhibit examples of hamiltonian in these algebra, having a discontinuous density of states. There are also models studied by physicists, like the laplacian on a Serpinsky lattice [1,85], which admit a discontinuous density of states. Thus we ask the following questions :

<u>Question 1</u> : What are the necessary and sufficient conditions on a self adjoint operator H affiliated to $C^*(\Omega,G)$ to get the convergence of the integrated density of states ?

Question 2 : What are the self adjoint operators H affiliated to $C^*(\Omega,G)$, having an integrated density of states continuous with respect to the energy, when G is amenable ?

Before giving a more precise result, let us introduce another tool. A G-invariant probability measure μ on Ω defines a trace on $C^*(\Omega,G)$ in the following way (where "e" is the neutral element in G) :

$$(3) \qquad \tau_\mu(a) = \int d\mu(\omega)\ a(\omega, e) \qquad a \in C_c(\Omega x G)$$

where $C_c(\Omega x G)$ is the space of continuous functions with compact support on $\Omega x G$. For indeed, it is a linear map, such that $\tau_\mu(a^*a) \geq 0$ and that $\tau_\mu(ab) = \tau_\mu(ba)$. It can be extended to a (weakly dense) ideal J of $C^*(\Omega,G)$, which is often denoted by $L^1(C^*(\Omega,G),\tau_\mu)$. By the Birkhoff ergodic theorem, if μ is G-ergodic, then for any Følner sequence we get :

$$(4) \qquad \tau_\mu(a) = \lim_{n\to\infty} |\Lambda_n|^{-1} \int_{\Lambda_n} dx\ a(x^{-1}.\omega, e) \qquad \text{for } \mu \text{ a.e. } \omega$$

Using the covariance condition , this is also equivalent to :

$$(5) \qquad \tau_\mu(a) = \lim_{n\to\infty} |\Lambda_n|^{-1} \mathrm{Tr}(\chi_{\Lambda_n} \pi_\omega(a)) \qquad \text{for } \mu \text{ a.e. } \omega$$

Hence such a trace is nothing but a **trace per unit volume**, and the limit converges for μ almost all ω. Let us note the following properties of the trace :

Proposition 6 : (i) If G is amenable, there is always at least one G-invariant ergodic probability measure μ on Ω.
(ii) For any G-invariant ergodic probability measure μ on Ω, and any $a \in C_c(\Omega x G)$, the limit (5) exists for μ almost all ω in Ω, and defines a trace on $C^*(\Omega,G)$.
(iii) If the support of μ is dense in Ω, this trace is faithful.
(iv) If G is discrete, this trace is bounded and normalized, i.e., $\tau_\mu(\mathbf{1}) = 1$ if $\mathbf{1}$ is the identity operator.
(v) If the dynamical system (Ω,G) is uniquely ergodic, namely if there is a unique G-invariant ergodic probability measure μ on Ω, then the limit in (5) is uniform with respect to ω.

◇

That (i) is true comes from a characterization of amenable groups by the existence of invariant means [M12]. Namely, on the C*algebra of continuous bounded functions on G, there is a G-invariant state M, if and only if G is amenable. Such a state is called an invariant mean. Therefore, if f is a continuous function on Ω, the function $x \in G \to f(x^{-1}.\omega) = F(x)$ is continuous and bounded on G, and the quantity $\mu(f) = M(F)$ is a positive linear map such that $\mu(1) = 1$. By Riesz's theorem, μ is a probability measure on Ω.

(ii) is just a consequence of Birkhoff's ergodic theorem [M13]. (iii) expresses the fact that any continuous positive function f on Ω such that $\mu(f) = 0$, vanishes μ-almost everywhere, and , by continuity, everywhere. The proof of (v) is a classical result in the study of dynamical systems (see [M5,M11,52]). ◇

The main interest of the trace, is to give a simple mathematical tool which describes what is called by physicists the **self averaging observables** namely, those physical quantities which are independent of the disorder. They are obtained through averaging a local observable over the sample. It has also been used, without notifying the formal definition of a trace, by R. Johnson & J. Moser [55], in dealing with the rotation number of a one dimensional Schrödinger equation (see section 6 below). The first non trivial result using this formalism was provided by L. Coburn, Moyer and I. Singer [24] who proposed to use C*algebras for extending the index theorem to pseudodifferential operators with almost periodic coefficients. This paper plaid a very important role in the original work of A. Connes when he generalized the index theorem for pseudodifferential operators on a foliated manifold, differentiating along the leaves. Following this proposal Shubin [93] in the mid seventies, studied the C*algebra of zeroth order pseudodifferential operators with almost periodic coefficients. He also studied the properties of the trace and realized that there is a connection between the integrated density of states and the trace, a result described below. Before giving this formula, we recall that if τ is a trace on a C*algebra $\mathcal{A}$, then through the GNS construction [M19], there is a representation π on the Hilbert space $L^2(\mathcal{A},\tau)$ (the completion of $\mathcal{A}$ under the Hilbert norm $\|a\|_2 = \{\tau(a^*a)\}^{1/2}$) such that if η is the canonical map from $\mathcal{A}$ into $L^2(\mathcal{A},\tau)$, we have :

$$(6) \qquad \tau(a^*bc) = \langle \eta(a) | \pi(b) \eta(c) \rangle \qquad a,b,c \in \mathcal{A}$$

We denote by $\mathcal{M}_\tau$ the W*algebra generated by $\pi(\mathcal{A})$ i.e. $\pi(\mathcal{A})''$. Then τ extends to $\mathcal{M}_\tau$. Moreover, if h is a self adjoint element of $\mathcal{A}$, then $\pi(h)$ has a spectral decomposition and the spectral projections belong to $\mathcal{M}_\tau$. Then we have :

<u>Shubin's formula</u> : Let H be a self adjoint operator on $L^2(G)$ affiliated to $C^*(\Omega,G)$, and μ be a G-invariant ergodic probability measure on Ω. We denote by $\chi(H \leq E)$ the spectral projection of H on energies smaller than or equal to E. Then, under one of the condition (Ai)

below, the density of states exists and for all E, and μ almost all ω one gets :

(7) $$\mathbf{N}_{\omega}(E) = \tau_{\mu}(\chi(H \leq E))$$

◇

From what we already explained, this formula is intuitively obvious (see eq. 2 & 5). However we need somewhere to control the restriction of H to a finite volume : as we said, if f is a Borel function on the spectrum of H, the restriction of f(H) is not equal to the image through f of the restriction of H This is hopefully true only asymptotically in the infinite volume limit. For this reason there is no general proof of this formula yet. However it is true under one of the conditions below :

(A1) G is discrete and H is bounded (see the appendix).

(A2) $G = \mathbf{R}^D$ and H is a pseudodifferential operator with quasi periodic coefficients (see Shubin [93], an extension of this work to the case where Ω is any smooth manifold should be easy))

(A3) $G = \mathbf{R}^D$ and H is a Schrödinger operator (see Bellissard, Lima, Testard [19])

The first case corresponds to the "lattice approximation", while the others correspond to the real continuum case. Treating the general case is the object of question 1.

5)- Cantor spectra, gap labelling theorem and K_0-group :

Using the covariance property of a homogeneous operator, one gets some elementary informations about its spectrum.

Proposition 7 : Let $(\Omega,\mathbf{R}^D,T)$ (resp. $(\Omega,\mathbf{Z}^D,T)$) be a dynamical system, and μ be a T-invariant ergodic probability measure on Ω. Let $H=\{H_\omega\,;\omega\in\Omega\}$ be a covariant μ-measurable family of self adjoint operators on a Hilbert space $L^2(\mathbf{R}^D)$ (resp. $l^2(\mathbf{Z}^D)$). Then :

(i) there is a closed subset Σ of $\mathbf{R}$ such that for μ-almost all ω in Ω, the spectrum of H_ω is Σ.

(ii) if E is an isolated point of Σ it is an eigenvalue of infinite multiplicity of H_ω, μ-almost surely.

(iii) if $\omega\to H_\omega$ is stongly continuous, Σ is the union of spectra of $\{H_\omega\,;\omega$ in the support of $\mu\}$.

(iv) if $\omega\to H_\omega$ is stongly continuous, and if Ω admits periodic points for T in the support of μ, then the set Σ cannot be nowhere dense.

◇

The proof of (i) and (ii) is classical [see 66]. For (iii)and (iv) it suffices to remark that if ω belongs to the support of μ, it may be approximated by a sequence of points ω' for which the spectrum of $H_{\omega'}$ is precisely Σ. Since the spectrum does not increase by strong limit [M18], Σ contains the spectrum of H_ω. If ω is a T-periodic point, H_ω admits a band spectrum [M7,M22,100]: it cannot be nowhere dense. ◇

The previous result applies in particular to the Schrödinger operators on $\mathbf{R}^D$ or $\mathbf{Z}^D$ with a random potential to show that the spectrum is actually connected [see 66]. In contrast, it has been shown that several examples of Schrödinger operators with almost periodic potential, have a nowhere dense spectrum. According to the previous result, this is impossible for a flow having a periodic orbit in the support of an invariant ergodic probability measure. However, this property is actually generic in one dimension for limit periodic potentials, as was shown firstly by J. Moser [75], and by J. Avron & B. Simon [11]. This is also true for a generic set of pairs (α,μ) for the almost Mathieu operator (see §3 eq.16) acting on $l^2(\mathbf{Z})$ as [18,see 23,53] :

$$(1)\qquad \psi(n+1)+\psi(n-1)+2\mu\cos 2\pi(\xi-n\alpha)\psi(n) = H\psi(n)\qquad n\in\mathbf{Z}$$

It is also believed to be generic for Schrödinger operator with an almost periodic potential. Cantor spectra have been found also for different examples of Jacobi matrices in one dimension. For instance if in (1) the cosine is replaced by a characteristic function of length α, numerical works indicate that this happens [63,79,80]. There is also a class of models [12,13,15,77], having the Julia set of a polynomial as spectrum.

Nobody gave any example of non almost periodic Schrödinger operator in one dimension having a Cantor spectrum, even though there is no reason "a priori" that Cantor spectrum is impossible for non almost periodic potentials. One interesting question in this respect is the following :

Question 3 : Let C be a Cantor set contained in a bounded interval of $\mathbf{R}$. Let μ be a probability measure on C the support of which being C itself. Let H be the operator of multiplication by x in $L^2(C,d\mu)$ and let $(p_n)_{n\geq 0}$ be the orthonormal basis generated by the Schmidt procedure from the set of monomials. Then the matrix of H in this basis is tridiagonal [M9]. When is this matrix almost periodic ?

Let us note however that the previous property is only generic. Non generic examples of one dimensional Schrödinger operators on $\mathbf{R}$ with finitely many bands in their spectrum have been constructed, using the inverse scattering method, in connection with the KdV equation, by Dobruvin, Matveev and Novikov [38].

The situation in more than one dimension is still unclear. There is only one known class of examples of Schrödinger operators, namely the discrete Laplace operators on "Serpinsky gaskets", which have a Cantor set as set of limit points of their spectrum. This was shown by R. Rammal [85, see 1]. However, for the periodic case, Skriganov [95] shown that in 2 dimensions, at high energy the spectrum is connected. We believe that this property survives even for a large class of homogeneous potential, because of the "band overlapping". However it may be perfectly possible that at low energy, there is some Cantor spectrum.

Nevertheless, the question remains in these cases to label the gaps, when the spectrum is a Cantor set. The solution to this problem is a question of taste : it depends whether one is a physicist or a mathematician. If one is a physicist, one will observe that the integrated density of states is locally constant outside the spectrum : it is constant on each gap (see fig 1 below). Since it is a strictly increasing function of the energy on the spectrum, two gaps separated by some non empty piece of the spectrum correspond to different values of the density of state. Thus a given gap J can be labelled by the value $\mathcal{N}(E)$ for $E \in J$.

A mathematician will not be very happy with this method, for it seems rather arbitrary. He would like some more canonical way. Thinking a little bit about the problem, one realizes that there is a mathematical object which can be affected to a gap J of the spectrum, namely the eigenprojection $P(J) = \chi(H \leq E)$ for any $E \in J$. Since E does not belong to the spectrum, the function $x \to \chi(x \leq E)$ is continuous on the spectrum and this eigenprojection

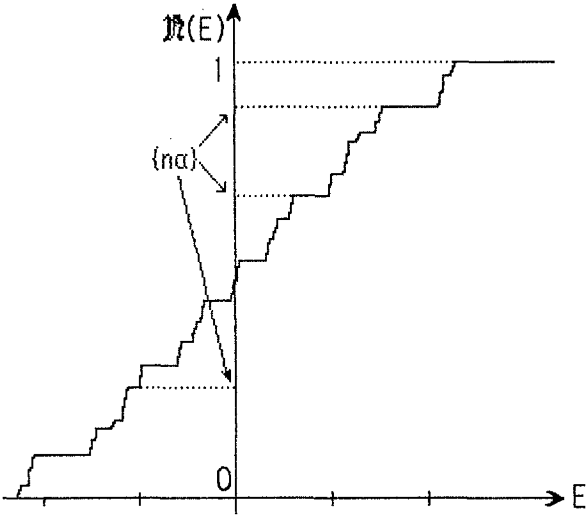

Fig. 1 : The integrated density of states for a one dimensional Schrödinger operator on a lattice or for any self adjoint bounded operator affiliated to the irrational rotation algebra.

still belongs to the C*algebra of the Hull ! This not a trivial property, for in a separable C* algebra projections are rather exceptional : for instance the projections of the C*algebra $\mathcal{K}$ of compact operators on a separable Hilbert space are all the finite dimensional ones. However giving the eigenprojection of the gap is certainly too much an information. For the spectrum does not change under unitary transformations. Thus only the equivalence class of such projections under unitary transformations is needed to label the gap. More precisely [M19,M25] :

<u>Definition</u> : Let $\mathcal{A}$ be a C* algebra with a unit, and P, Q be two projections in $\mathcal{A}$. Then P and Q are equivalent, and we will denote this relation by $P \approx Q$, if there are two elements U and V in $\mathcal{A}$ such that :

(2) $\quad P = UV \qquad Q = VU$

If P and Q are self adjoint then one can take $V = U^*$. ◇

One could ask the question why we have chosen such an equivalence relation. For indeed as far as the spectrum as a set is concerned, any automorphism of the C*algebra would leave it invariant. Identifying two projections obtained one from the other through such an automorphism would diminish the set of equivalence classes in general. Our choice is motivated by the remark that unitary transformations are those automorphisms which are approximated by local automorphisms, when H is approximated by its restrictions on bounded sets.

The next problem is to compute the set $\mathcal{P}$ of equivalence classes. Actually nobody knows how to do that in general ! However as remarked by

Grothendieck [M3], there is some structure in this set provided one accepts to weaken a little bit our notion of equivalence. For indeed, if [P] denotes the equivalence class of the projection P, and if P,Q are orthogonal to each other (namely $PQ = QP = 0$) then $P+Q = P\oplus Q$ is also a projection the equivalence class of which depending only upon [P] and [Q]. Therefore we get an addition in $\mathbf{P}$ defined by :

(3) $\quad [P] + [Q] = [P'\oplus Q'] \quad$ for any $P'\approx P$ and $Q'\approx Q$ with $P'Q = Q'P = 0$

Again this addition is not defined for any pair of projections, for it may happen that there is not enough room in $\mathcal{A}$ to rotate this pair in such a way that the two projections become orthogonal. However, if we enlarge $\mathcal{A}$ by adding the union over $n \geq 0$, of the $n\times n$ matrices with entries in $\mathcal{A}$, this will work. Indeed, we will identify P with the matrix

$$(4) \qquad P \approx \begin{vmatrix} P & 0 \\ 0 & 0 \end{vmatrix} \approx \begin{vmatrix} 0 & 0 \\ 0 & P \end{vmatrix} \qquad \text{in } \mathcal{A}\otimes M_2$$

Hence in $\mathcal{A}\otimes M_2$ it is always possible to transform P and Q in $\mathcal{A}$ into a pair of equivalent orthogonal projections. However this will not be true for any pair in $\mathcal{A}\otimes M_2$. Thus we will replace $\mathcal{A}\otimes M_2$ by $\mathcal{A}\otimes M_2\otimes M_2$ and if we keep going we eventually end up with the algebra $\mathcal{A}\otimes\mathcal{K}$. Now this is enough for $\mathcal{K}\otimes\mathcal{K}$ is isomorphic to $\mathcal{K}$ and therefore $\mathcal{A}\otimes\mathcal{K}$ is isomorphic to $\mathcal{A}\otimes\mathcal{K}\otimes\mathcal{K}$. For this reason $\mathcal{A}\otimes\mathcal{K}$ is called the **stabilized algebra**. To conclude, the set $\mathbf{P}$ of equivalence classes of projections in the stabilized algebra is endowed with an addition which is commutative. Then by "abstract nonsense", Grothendieck told us that there is a canonical group which can be constructed out of $\mathbf{P}$ exactly like we construct $\mathbf{Z}$ out of $\mathbf{N}$: this group called $K_0(\mathcal{A})$, is the set of classes of pairs ([P],[Q]) where the equivalence is given by :

(5) $\quad \exists\, [R]\in P \quad [P] + [Q'] + [R] = [P'] + [Q] + [R] \quad \Leftrightarrow \quad ([P],[Q]) \approx ([P'],[Q'])$

Then [P]-[Q] is identified with the class of the pair ([P],[Q]), and the addition extends to a group law. Moreover let us note the following result :

<u>Proposition 8</u> [M19] : If $\mathcal{A}$ is a separable C*algebra, the group $K_0(\mathcal{A})$ is abelian and countable ◇

Now we remark that any trace τ on $\mathcal{A}$ satisfies the two following properties :

(6) (i) if $P \approx Q$ then $\tau(P) = \tau(Q)$
(ii) if $PQ = QP = 0$ then $\tau(P \oplus Q) = \tau(P) + \tau(Q)$

This shows that τ defines a map τ_* on $K_0(\mathcal{A})$ which is actually a real valued homomorphism through the formula :

(7) $$\tau_*([P]) = \tau(P)$$

Moreover if τ is faithful (namely $a \geq 0$ and $a \neq 0$ implies $\tau(a) > 0$) then τ_* is one-to-one.

Coming back to our original problem we see that if J is a gap the number $\tau(P(J))$ which, by Shubin's formula, is identical to the value of the integrated density of states on the gap, is equal to the number $\tau_*([P])$ and therefore :

Gap labelling theorem 1 : Let H be a self adjoint operator affiliated to $C^*(\Omega,G)$, and such that its density of states exists with respect to a G-invariant ergodic probability measure μ on Ω . Then the values of the density of states on a gap of the spectrum belongs to the positive part of the countable subgroup of $\mathbf{R}$ given by $\tau_{\mu*}(K_0(C^*(\Omega,G)))$. ◇

Corollary 1(homotopy invariance) : Let H be as before, let J be a gap in the spectrum of H, and let $\tau(J)$ be the value of the integrated density of states on J. Then $\tau(J)$ is invariant under small perturbations of H in the norm resolvent convergence. ◇

Even though this construction may appear rather complicate, it is actually very natural. For instance, the **stabilized algebra of a dynamical system is isomorphic to the algebra of its suspension** [29,30,87]! Thus stabilizing is nothing more than the **inverse operation** described in the section 3, namely the **tight binding representation** .

The obvious question now is how to compute the K-group ! Precisely this was one of the breakthrough in the early eighties, to give a calculation of these groups. The first important result was given in 1979 by Pimsner and Voiculescu [81,82] and it concerns the irrational rotation algebra :

Theorem 1 : If α is an irrational number, let $\mathcal{A}_\alpha$ be the algebra generated by two unitaries U and V such that $UV = e^{2i\pi\alpha} VU$, (i.e. $\mathcal{A}_\alpha = C^*(\mathbf{T},\mathbf{Z})$, where $\mathbf{Z}$ acts on $\mathbf{T}$ through the rotation R_α). Then :
(i) on $\mathcal{A}_\alpha$ there is a unique trace τ which is faithful (prop.6).
(ii) the group $K_0(\mathcal{A}_\alpha)$ is isomorphic to $\mathbf{Z}^2$. Its generators are given

by the equivalence classes of the identity $\mathbf{1}$, and of the "Rieffel projection" P_R [86].

(iii) The subgroup $\tau_*(K_0(\mathcal{A}_\alpha))$ is equal to $\mathbf{Z}+\alpha\mathbf{Z}$.

(iv) If h is any self adjoint element of $\mathcal{A}_\alpha$, and J is a gap in its spectrum, there is a unique $n \in \mathbf{Z}$ such that the integrated density of states takes on the value $n\alpha-[n\alpha]$ on J. ◊

The previous argument leading to the gap labelling theorem1, was presented in a meeting at Marseille in April 1981, and announced in the IAMS conference in Berlin in August 1981[14]. On the other hand it can also been found in the review by B. Simon [94] on almost periodic Schrödinger operators. This result was also proven in a direct way by Delyon and Souillard [35] in the very special case of the almost Mathieu operator (with an arbitrary continuous function V instead of the cosine, as a potential in (1)). They actually used a suspension technics together with the work of Johnson and Moser [55] who gave a version of the gap labelling theorem for the one dimensional Schrödinger operator on $\mathbf{R}$ (see below).

Let us also mention that the result of Pimsner and Voiculescu is more general in that they actually computed the K-group for a wider algebra, which is generated by the translation operator U in $l^2(\mathbf{Z})$ together with the operator χ of multiplication by $\chi_{(0,\alpha)}(x-n\alpha)$. Since this new algebra contains the irrational rotation algebra, what they proved was that the K-group of the latter was not bigger than $\mathbf{Z}^2$. To prove that it was equal, they used the first work of M. Rieffel [86] who produced an example of projection P_R, in the form of a tridiagonal matrix, the trace of which was equal to α. Nevertheless this extension of theorem 1 is important to notice for the Schrödinger operator (1) with $\mu\chi$ as a potential became popular recently because it represents the operator generating the **phonon spectrum of a one dimensional quasi crystal**.

However, even though important, the previous theorem was just a first step toward a general theory. The next breakthrough was performed by A. Connes few weeks after he received the work by Pimsner and Voiculescu, and generalized it in giving a geometrical interpretation together with explicit formulæ to compute the image of the K-group by the trace homomorphism. Actually, as soon as 1977, A. Connes had proved a generalization of the index theorem for pseudodifferential operators on a foliated manifold [26,27,31], and this earlier result contained in essence what we are going to explain now.

<u>Definition</u> : Let G be a Lie group of dimension D, acting freely and smoothly on a smooth compact manifold Ω. Let $X_1,\dots,X_D$, be vector fields generating the G-action on Ω. Let also μ be a G-invariant ergodic probability measure on Ω (if G is not amenable μ may or may not exist). The Ruelle-Sullivan current [88] is defined as the linear

form on the space of D-differential forms on Ω as follows :

(8) $$\langle \mathfrak{J} \mid \eta \rangle = \int_{\Omega} d\mu \langle \eta \mid X_1 \wedge \ldots \wedge X_D \rangle$$

◇

Then $\mathfrak{J}$ is closed and is positive (i.e. $\langle \mathfrak{J} \mid \eta \rangle \geq 0$ if η is positive on each G-orbit) and this is a characterization of the Ruelle-Sullivan current [30]. To say that $\mathfrak{J}$ is closed means that it is zero on exact forms $\eta = d\varphi$ (φ a (D-1)-differential form) and this is equivalent to say that μ is invariant by G. Therefore $\mathfrak{J}$ defines a linear map denoted by $[\mathfrak{J}]$, from the D-cohomology group $H^D(\Omega,\mathbf{R})$ into $\mathbf{R}$.

Now the main result by A. Connes [30] in our context is the following :

Gap labelling theorem 2 : Let Ω,G as before. Then the countable subgroup of $\mathbf{R}$ given by the image under the trace homomorphism of the K_0-group of $C^*(\Omega,G)$ is equal to the image of the integer D-cohomology group $H^D(\Omega,\mathbf{Z})$ by the homology class $[\mathfrak{J}]$ of the Ruelle-Sullivan current.

◇

In practice this means that we must compute a set of D independent D-cycles in Ω, generating the homology group $H^D(\Omega,\mathbf{R})$. Then, by duality, we must exhibit a set of D linearly independent closed and non cohomologous differential forms of degree D, the integrals of which on the previous cycles being integers. Then we evaluate $\mathfrak{J}$ on this last set of form using (8). The subgroup of $\mathbf{R}$ generated by the numbers that we obtain in this way gives the answer.

Let us consider the special case where Ω is a torus of dimension $\nu > D$, and $\mathbf{R}^D$ acts on it via constant vector fields $\alpha_1,\ldots,\alpha_D$. Let α be the $\nu \times D$ matrix the columns of which being given by the coordinates of the α_i's. To get a free action, this matrix must be "irrational", namely there is no non zero vector m in $\mathbf{R}^D$ with integer coordinates, such that αm be a vector with integer coordinates. Then the dynamical system is an abelian virtual group, and it corresponds to the hull of some Schrödinger operator on $\mathbf{R}^D$ with a quasi periodic potential. We know explicitly a set of generating D-forms on $\mathbf{T}^\nu$, namely $d\omega_{i_1} \wedge \ldots \wedge d\omega_{i_D}$ where the $d\omega_i$'s are the differential of the coordinate functions. There is also a unique invariant measure in this case, namely the Haar measure $d\omega_1 \wedge \ldots \wedge d\omega_\nu$ on the ν-torus. The evaluation of (8) becomes elementary and this gives :

Corollary 1 [19] : In the quasi periodic previous case, the integrated density of states of a self adjoint operator H affiliated to $C^*(\mathbf{T}^\nu, \mathbf{R}^D)$ takes on positive values on the gap of H belonging to

(9) $$L = \Sigma_{(i)} \mathbf{Z} \alpha^{(i)}$$

where the $\alpha^{(i)}$'s are the minors of maximal rank of the matrix α.

◇

A special case is given by D=1. Then α is a column matrix, namely a vector in $\mathbf{R}^\nu$, and the minors of maximal rank are just the coordinates of this vector. A typical potential admitting this dynamical system as a Hull, has the form :

(10) $$V(x) = v(\omega-\alpha x) = \Sigma_{m \in \mathbf{Z}^\nu} e^{2i\pi(\omega-x\alpha)m} v(m) \qquad x \in \mathbf{R}$$

where the right hand side represents the Fourier expansion of $v \in C(\mathbf{T}^\nu)$. Thus the coordinates of α represents the independent frequencies of V. The "frequency module" is defined as the group generated by the elementary frequencies, and we get as a corollary [see 55] :

Corollary 2 : (i) Let $H = -d^2/dx^2 + V$ be a Schrödinger operator on $\mathbf{R}$ with a quasi periodic potential V. Then the integrated density of states on a gap takes on values in the frequency module of V.
(ii) If H is any self adjoint pseudodifferential operator on $\mathbf{R}$ with quasi-periodic coefficients, the same result holds, with the frequency module of the coefficients replacing the one of the potential.

◇

The result (i) was first obtained in 1981 by Johnson and Moser [55, see 54], who used all the properties of a second order differential operator to prove it. The other result (ii) is a consequence of Shubin's analysis, and of the gap labelling theorem 2.

The virtue of our approach is that it extends as well to the tight binding approximation :

Proposition 8 : (i) Let (Ω,G) be a dynamical system, where G is connected, and let Ξ be a transversal. Then the stabilized algebras of $C^*(\Omega,G)$ and of $C^*(\Xi)$ are isomorphic (one says that they are Morita equivalent). Consequently their K-group are isomorphic.
(ii) If $G=\mathbf{Z}^D$, there is a compact space Ω' together with a $\mathbf{R}^D$-action such that $(\Omega,\mathbf{Z}^D)$ is isomorphic to the dynamical system of a transversal of Ω'. The system $(\Omega',\mathbf{R}^D)$ is called the suspension of $(\Omega,\mathbf{Z}^D)$. ◇

The proofs of this result are scattered in the literature. The concept of Morita equivalence is quite old [see 87], and the connection with the C*algebra of a transversal is the result of works by M. Rieffel [87], and A. Connes [30]. The suspension construction was well known from the experts in dynamical systems for a long time but we can find an exposition of it in [72(remark),96]. As a consequence we get [see also 42]:

Corollary 3 [19] : (i) Let H be a self adjoint operator on $\mathbf{Z}^D$ affiliated to the C*algebra of $(\mathbf{T}^\nu,\mathbf{Z}^D)$ where $m \in \mathbf{Z}^D$ acts on $\mathbf{T}^\nu$ via the translation by αm where α is a $\nu \times D$ matrix the minors of which being rationally independent. Then the integrated density of states of H on a gap takes on values in the set of linear combinations with integer coefficients of the minors of α of any order (with the convention that 1 is a minor of zeroth order).
(ii) Let H be the discrete laplacian on $\mathbf{Z}^D$ with a quasi periodic potential V, the hull of which being the previous dynamical system. Then (i) apply to it. In particular if D=1, the integrated density of states of H on a gap takes on values in the set $\mathbf{Z} + L$ where L is the frequency module of V

◇

To finish with this section let us give few more results which were noticed in [14,17]. As we noticed in proposition 7, to get a nowhere dense spectrum we need flows without periodic orbits. This is actually not sufficient. It is also necessary that the image of the K-group by the trace be dense in $\mathbf{R}$. Let us mention one example, noticed by A. Connes, of flow for which this does not happen. A. Connes [29] gave a series of sufficient condition for a discrete dynamical system $(\Omega,\mathbf{Z})$ where Ω is a manifold, to insure that $C^*(\Omega,\mathbf{Z})$ is simple without projection. In this case, obviously, the spectrum of any self adjoint element of $C^*(\Omega,\mathbf{Z})$ has no gap.

Example 1 : let Γ be a discrete subgroup of $SL(2,\mathbf{Z})$ with a compact fundamental domain, and let Ω by the compact space $SL(2,\mathbf{R})/\Gamma$. Let φ be a minimal diffeomorphism of Ω (namely such that any orbit is dense). The horocycle flow provides such an example. Then the K_0-group for this discrete flow is equal to $\mathbf{Z}$. In particular since the flow is minimal, its C*algebra is simple [29,43], and any trace τ is faithful. Therefore there is no non trivial projection otherwise there would be a non zero projection P with $\tau(P) \in \mathbf{Z}$. Since $\mathbf{Z}$ is discrete and Ω compact, τ is normalized, and we must have $0 \leq \tau(P) \leq 1$. Thus $\tau(P) = 1$ and $\tau(\mathbf{1}-P) = 0$ which implies $P=\mathbf{1}$. Let now v be a continuous function on Ω, and H be the operator on $l^2(\mathbf{Z})$ given by :

$$H\psi(n) = \psi(n+1) + \psi(n-1) + v(\varphi^{-n}\omega)\,\psi(n) \qquad (11)$$

Then H has no gap in its spectrum. ◇

The next example [17] goes just in the opposite direction:

<u>Example 2</u>: let us consider the Schrödinger operator on $l^2(\mathbf{Z})$:

$$(12) \qquad H\psi(n) = \psi(n+1) + \psi(n-1) + \lambda\chi_{(0,\beta]}(x-n\alpha)\,\psi(n)$$

where $1,\alpha,\beta$ are rationally independent. It was shown in [17] that the gap labelling theorem was different from what was obtained from the Pimsner Voiculescu theorem. The reason is that in the corresponding C*algebra of the Hull, there is a new projection, namely $\chi_{(0,\beta]}$ the trace of which being β. For this reason the K-group is strictly wider and its image by the trace contains $\mathbf{Z} + \mathbf{Z}\alpha + \mathbf{Z}\beta$. The same phenomena would appear if instead of taking the irrational rotation by α we chose a Denjoy diffeomorphism of the circle [M14,84].

Let us note the last problem :

<u>Question 4</u> : Thank to the gap labelling theorem, a necessary condition for a self adjoint operator H, affiliated to $C^*(\Omega,G)$ to have a Cantor spectrum is that the image of the K-group by the trace homomorphism is dense in the real line. Another necessary condition is that Ω be free of periodic orbits. Let us suppose that these two conditions hold. Has a generic self adjoint element of $C^*(\Omega,G)$ a Cantor spectrum ?

6)- The case of flows : Johnson's approach.

When investigating the properties of one dimensional Schrödinger operators with random potential, R. Johnson [56] gave another version of the gap labelling theorem, using the notion of rotation number already described by R. Johnson & J. Moser [55], and by M. Herman [52]. He actually found an explicit formula which turned out to be nothing but the Connes formula given in §5 eq.8. However, being more precise, this approach allows us to understand the origin of the "Ruelle-Sullivan" current.

Let us consider a compact metric space Ω, together with a continuous flow $t\in\mathbb{R}\to T_t\omega$, and let μ be an ergodic invariant measure on it. Let now $\omega\to M(\omega)$ be a continuous map on Ω with values in the set of 2x2 matrices with real entries. We now consider the ordinary differential equation :

$$(1) \qquad dX/dt = M(T_{-t}\omega)X(t) \qquad X(t)\in\mathbb{R}^2$$

Without loss of generality we may assume that $M(\omega)$ is traceless for all ω. The solution of (1) can always be written as :

$$(2) \qquad X(t) = \Phi(\omega,t)\,X(0)$$

where $\Phi(\omega,t)$ is a 2x2 real matrix with determinant one satisfying the cocycle properties :

$$(3) \qquad \Phi(\omega,0) = \mathbf{1} \qquad \Phi(\omega,t+s) = \Phi(T_{-s}\omega,t)\,\Phi(\omega,s)$$

We will say that (1) admits an **exponential dichotomy** [89,90] if the trivial fiber bundle $\Omega\times\mathbb{R}^2$ splits into a direct sum of non zero subbundles $E^+\oplus E^-$ such that :

(i) if $(\omega,X)\in E^\pm$, then $(T_{-t}\omega, \Phi(\omega,t)X)\in E^\pm$ for all $t\in\mathbb{R}$ (invariance of $E^\pm$)

(ii) there are positive constant K and r independent of (ω,t) such that if $(\omega,X)\in E^\pm$ then $\|\Phi(\omega,t)X\| \le K e^{\pm(-rt)}$

An example of such flow is provided by a one dimensional Schrödinger equation (where V is a continuous function on Ω) :

$$(4) \qquad -d^2\psi/dt^2 + V(T_{-t}\omega)\psi(t) = E\,\psi(t)$$

where we set :

$$(5) \qquad X(t) = \begin{bmatrix} X_1(t) = \psi(t) \\ X_2(t) = \psi'(t)/\sqrt{E} \end{bmatrix} \qquad M(\omega) = \begin{vmatrix} 0 & \sqrt{E} \\ -\sqrt{E}+V(\omega)/\sqrt{E} & 0 \end{vmatrix}$$

It admits an exponential dichotomy provided E does not belong to the spectrum of the self adjoint operator described formally by (4) on $L^2(\mathbf{R})$ [56,90]. More precisely, for each $\omega \in \Omega$ there is a solution $u^+(\omega,t)$ (resp. $u^-(\omega,t)$) of (4) unique up to a multiplicative constant, decreasing exponentially at $+\infty$ (resp. $-\infty$) together with its first and its second derivative.

Let $X^\pm(\omega,t)$ be solutions of (1) in $E^\pm$, written in the form :

$$(6) \qquad X^\pm(\omega,t) = r^\pm(\omega,t)\, \mathbf{e}(\theta^\pm(\omega,t))$$

where $\mathbf{e}(\theta)$ is the unit vector of components $\cos\theta$ and $\sin\theta$. We remark that the line passing through $X^\pm$ is uniquely determined by $\theta^\pm$ (modulo π). Thanks to the uniqueness of $X^\pm$, it is easy to see that the angle $\theta^\pm(\omega,t)$ is actually a function of $T_{-t}\omega$ only, which we will denote by $\theta^\pm$ again.

The rotation number of this solution is defined as the amount of angle per unit length namely :

$$(7) \qquad \rho^\pm = \lim_{t\to\pm\infty} \{\theta^\pm(T_{-t}\omega) - \theta^\pm(\omega)\}/t$$

Let us assume that Ω is a manifold and that the flow T on it is defined through the vector field ξ. By using the Birkhoff ergodic theorem, and provided $\theta^\pm$ is differentiable in ω, we get, for μ-almost every ω_0 :

$$(8) \quad \rho^\pm = \lim_{t\to\pm\infty} (1/t)\int_0^t ds\ \partial\theta^\pm(T_{-s}\omega_0)/\partial s$$
$$= \int d\mu(\omega)\, \langle d\theta^\pm | \xi\rangle(\omega) = \langle \mathfrak{J} \mid d\theta^\pm \rangle$$

where $\mathfrak{J}$ is precisely the Ruelle-Sullivan current for the dynamical system. Let us remark that $d\theta^\pm$ is closed but not exact in general for $\theta^\pm$ is defined modulo π, and in particular, its integral on any loop (or 1-cycle) in Ω is an integer multiple of π. Therefore the cohomology class $[d\theta^\pm/\pi]$ belongs to $H^1(\Omega,\mathbf{Z})$. Thus we get :

<u>Theorem 2</u> (R. Johnson) : (i) The rotation numbers of the flow (1) satisfies $\rho^+=\rho^-$, and $\rho^\pm/\pi$ belongs to the image of the integer cohomology group $H^1(\Omega,\mathbf{Z})$ by the homology class of the Ruelle-Sullivan current. Moreover it is a homotopy invariant.

(ii) If the flow (1) comes from the Schrödinger equation (4), where E belongs to a gap of the spectrum, then the rotation number $\rho^\pm/\pi$ is equal to the integrated density of states and satisfies :

$$(9) \qquad \mathcal{N}(E) = \rho^\pm/\pi = \sqrt{E}/\pi \int d\mu(\omega)\ \{1 + \cos^2(\theta^\pm(\omega))\ V(\omega)/E\}$$

◇

Sketch of the proof : The first part is a consequence of the previous reasoning. The second is an application of the Sturm theorem : the number of zeroes of a solution of (4) in a finite interval $[-L,L]$ with some boundary conditions and the number of the corresponding eigenvalues smaller than or equal to E, differ at most by 2. On the other hand the number of zeroes of $\psi^{\pm}$ in $[-L,L]$ differs from $\{\theta^{\pm}(T_{-L}\omega)-\theta^{\pm}(T_L\omega)\}/\pi$ by at most 1. Dividing by 2L and letting L go to infinity gives the identity between the rotation number divided by π and the integrated density of states. From (5) we get :

$$(10) \qquad d\theta^{\pm}/dt = \sqrt{E}\,\{1 + \cos^2(\theta^{\pm})\, V(T_{-t}\omega)/E\,\} = \langle d\theta^{\pm}|\xi\rangle(T_{-t}\omega)$$

which gives the last formula. ◇

The previous analysis for flows extends to the case of discrete maps, namely :

$$(11) \qquad X(n+1) = M(T^{-n}\omega)\, X(n) \qquad n\in\mathbf{N}$$

The hypothesis will be now the following: M is a 2x2 real matrix valued continuous function with determinant one which is homotopic to the identity matrix. We then replace the discrete flow (Ω,T) by its suspension $(S\Omega,ST)$ where $S\Omega$ is the quotient of $\Omega\times\mathbf{R}$ by the equivalence relation $(\omega,t)\approx(T\omega,t-1)$, and ST is the quotient of the flow $(\omega,s)\rightarrow(\omega,t+s)$. In much the same way, we replace the discrete flow $(\omega,X)\rightarrow(T^{-1}\omega,M(\omega)X)$ on the trivial bundle $\Omega\times\mathbf{R}^2$ by its suspension. Since M is homotopic ti the identity, $S(\Omega\times\mathbf{R}^2)$ is homeomorphic to $S\Omega\times\mathbf{R}^2$. If (Ω,T) admits an exponential dichotomy, the suspension admits also an exponential dichotomy. Thus the previous analysis goes through. However we can now define the rotation number directly from (11) as was proposed by M. Herman [52], namely : any matrix in $SL(2,\mathbf{R})$ defines a diffeomorphism on the projective space $P^1(\mathbf{R})$, the set of lines in $\mathbf{R}^2$ which can be identified with the unit circle or with the torus $\mathbf{T}=\mathbf{R}/\mathbf{Z}$. Thus we get a mapping $(\omega,\theta)\in\Omega\times\mathbf{T}\rightarrow f(\omega,\theta)\in\mathbf{T}$ which is continuous and such that $f_\omega:\theta\in\mathbf{T}\rightarrow f(\omega,\theta)\in\mathbf{T}$ is a diffeomorphism of $\mathbf{T}$. Let us denote by f again the lifting of f on $\Omega\times\mathbf{R}$. It satisfies : $f(\omega,x+1) = f(\omega,x)+1$ $(x\in\mathbf{R})$. The rotation number is then defined as :

$$(12) \qquad \rho = \pi \lim_{n\rightarrow\infty} \{f_{T^{-n+1}\omega}\circ\ldots\circ f_\omega(x) - x\}/n$$

That this limit exists form μ-almost every ω and uniformly in x was one of the results of [52]. It is then easy to see that this rotation number is the same as the one defined through the suspension.

Again also, this approach can be used to investigate a one dimensional discrete Schrödinger operator of the form :

$$(13) \qquad H\psi(n) = t(T^{-n-1}\omega)\psi(n+1) + t(T^{-n}\omega)\psi(n-1) + v(T^{-n}\omega)\psi(n) = E\psi(n)$$

where v,t are continuous real functions on Ω and t is positive (actually it is sufficient that t be complex and non vanishing). Then the equation $H\psi = E\psi$ is equivalent to (11) with :

$$(14) \qquad M(\omega) = \begin{vmatrix} (E-v(\omega))/t(T^{-1}\omega) & t(\omega)/t(T^{-1}\omega) \\ 1 & 0 \end{vmatrix}$$

In this case also the density of states coincides with ρ/π. The results in examples are therefore identical to those given in the previous chapter.

Let us finish this section by describing an application of this framework to some problem in classical mechanics.

Let F be a monotone twist mapping of the annulus $A=\mathbf{T}\times[01]$, namely an homeomorphism which preserves the ends of A, $\mathbf{T}\times\{0\}$ and $\mathbf{T}\times\{1\}$, which is area preserving and such that if $y < y'$ then $F(x,y) < F(x,y')$. The restrictions of F to the ends define two diffeomorphism of the circle the rotation numbers of which will be called $\rho^{(0)}$ and $\rho^{(1)}$. It follows that $\rho^{(0)} < \rho^{(1)}$. One example is given by the "standard map" :

$$(15) \qquad F(x,y) = (x+y+k\sin 2\pi x,\ y+k\sin 2\pi x)$$

The Aubry-Mather theorem [7,8,57,58,73] shows that for any irrational ρ in the interval $(\rho^{(0)},\rho^{(1)})$, there is a closed invariant subset $M(\rho)$ such that the restriction of F on it is conjugate through a homeomorphism, to the restriction of a diffeomorphism f of the circle of rotation number ρ to its (unique) minimal invariant set. For indeed [see M14], any diffeomorphism f of $\mathbf{T}$ has a unique minimal invariant set which is the full circle if it is smooth enough (depending on ρ), whereas it is a Cantor subset of $\mathbf{T}$ otherwise, as was firstly shown by Denjoy [37]. Thus $M(\rho)$ is homeomorphic either to a circle or to a Denjoy set. If ρ is rational the situation is more involved [see 73].

Let F denote again the lifting of F on $B=\mathbf{R}\times[0,1]$. Since F is area preserving, using the monotone twist property it can be shown [73] that there is a C^1 periodic function $h(x,x')$ on $\mathbf{R}^2$, called the generating function of F, such that $y'=\partial h/\partial x'(x,x')$ and $y=-\partial h/\partial x(x,x')$. If we set $(x_n,y_n) = F^n(x,y)$ the sequence $\{x_n\}$ satisfies the following non linear equation :

$$(16) \qquad \partial h/\partial x'(x_{n-1},x_n) + \partial h/\partial x(x_n,x_{n+1}) = 0$$

which gives for the standard map :

$$(17) \qquad x_{n+1} + x_{n-1} - 2x_n + k\sin 2\pi x_n = 0$$

If now $(x,y) \in M(\rho)$ the Aubry-Mather theorem means that there are f,g the lifting of two homeomorphisms of $\mathbf{T}$, and $\xi \in \Sigma$, the minimal set of f, such that $x_n = g(f^n(\xi))$ and that the rotation number of f is ρ.

The linear stability of $M(\rho)$ is described through the linear equation governing the evolution of an infinitesimal change $\delta x_n = \psi(n)$ in x_n, namely, if h is in C^2, through the equation (13) with $\Omega = \Sigma$, $T = f$, E=0 and :

$$(18) \qquad t(\xi) = \partial^2 h/\partial x \partial x'(g \circ f^{-1}(\xi), g(\xi))$$

$$v(\xi) = \partial^2 h/\partial x'^2(g \circ f^{-1}(\xi), g(\xi)) + \partial^2 h/\partial x^2(g(\xi), g \circ f(\xi))$$

The rotation number of this discrete system is called here the amount of rotations. It has been studied by J. Mather [74] in the special case where ρ is a rational number and $\{x_n\}$ is periodic, where it is connected to the Morse index. He found a result which can be interpreted in term of K-theory. Let us give the result in the case where now ρ is irrational.

If E=0 does not belong to the spectrum of H, a property which is likely to be generic if one believes that the spectrum of H is a Cantor set, then the amount of rotation is nothing but the integrated density of states (if one normalizes the angle to 1 instead of π). Therefore the gap labelling theorem 1 shows that it belongs to the image by the (unique) trace on $C^*(\Sigma,f)$ of the K_0 group of this algebra. The uniqueness of the trace comes from the uniqueness of an f-invariant measure on $\mathbf{T}$ [M14]. The K-group of this algebra has been computed by I. Putnam, K. Schmidt & C. Skau [84]. If μ is the unique f-invariant probability measure on $\mathbf{T}$, μ is actually supported by Σ. If Σ is a Cantor set then $\mathbf{T} \backslash \Sigma$ is the union of at most countably many intervals which are the "gaps" of Σ. Then Q(f) is the set of values of $g(\xi) = \int_0^\xi d\mu(\eta)$ when ξ varies through the gaps of Σ. Clearly Q(f) is countable, and it is known that if $R(\rho)$ denotes the rotation by ρ, then $g \circ f = R(\rho) \circ g$, showing that Q(f) is also $R(\rho)$ invariant in $\mathbf{T}$ identified with [0,1). Therefore there is a family of real numbers $\{\gamma(i);\ 1 \leq i \leq n(f)\}$ (where n(f) may be infinite), such that Q(f) equals the union of the $\{\gamma(i)+\mathbf{Z}\rho\}$'s. Actually the γ's are uniquely defined if we assume that the differences $\gamma(i)-\gamma(j)$ are rationally independent of ρ. Moreover, Q(f) is defined up to a solid rotation. Therefore we may assume $\gamma(1)=0$. Then the result of the analysis of [84] is the following :

<u>Proposition 9</u> : The C^*-algebra $C^*(\Sigma,f)$ is simple and has a unique trace. The trace of any projection in it belongs to the set :

$$(19) \qquad \{\mathbf{Z} + \mathbf{Z}\rho + \mathbf{Z}\gamma(2) + \ldots + \mathbf{Z}\gamma(n(f))\} \cap [0,1]$$

If 0 does not belong to the spectrum of the linearisation H(F), the amount of rotation of the map F on the Cantorus $M(\rho)$ belongs also to this set. ◇

7)- The quantum Hall effect :

The Hall effect was discovered and discussed 1879 by E.H. Hall [49], and has been one of the most striking and useful phenomena in solid state physics for a century, since it was one of the first effect giving informations on the microscopical properties of metals. For instance it gives the sign of the charge of the particles carrying the current. This was the way one discovered that the electric current can be carried either by electrons or by holes depending on the material used. It gives also a method to measure the velocity of the carriers [21] and the charge carriers density. On the other hand, it is nowadays commonly used in several kind of devices in electronic, computer, electronic typewriters, automobile engines, etc.

In 1980, a century after the discovery of this effect, a new breakthrough was performed with the discovery [62] of its quantum counterpart which leads to such a precise experiment that the standard for the Ohm, the unit of resistance, is to be defined from it quite soon. Undoubtely, there will be other applications of this effect when the technology will be controlled in order to produce cheap devices.

Let us briefly recall the nature of the classical Hall effect, in order to go step by step toward the understanding of the quantum one. If a thin strip of a metal is submitted to a uniform magnetic field perpendicular to it, one observes a potential difference in the direction perpendicular to the electric current and the magnetic field (see fig. 2). Measuring this voltage leads to the value of the Hall resistance given by the ratio of this Hall voltage and the intensity of the current. To compute it, let us assume that the permanent regime is established. Two forces act on the unit volume of charged particle.

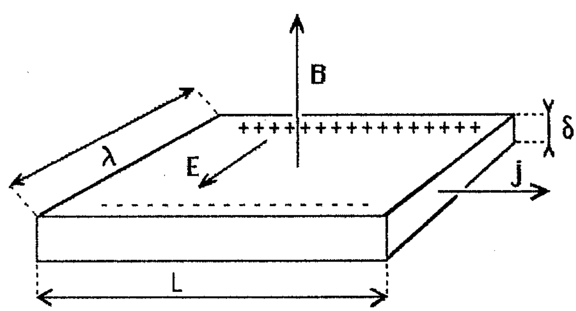

Fig. 2 : The classical Hall effect. The sample must be a thin strip of pure metal. The magnetic field B is perpendicular to the strip. The electric current j is stationary and oriented along the strip in the direction of the longest side. The magnetic forces push the charges as indicated. In the stationary state, they create an electric field E perpendicular to the directions of the magnetic field and the current. One measures the Hall voltage in the E-direction.

The first one is the Lorentz force, perpendicular to the magnetic field and to the current density. Its intensity is given by :

(1) $$F_{magn.} = jB/c$$

Here c is the speed of light, j is the current density namely the ratio $I/\lambda\delta$, where I is the intensity of the current, δ the thickness and λ the width of the strip. B denotes the magnitude of the magnetic field in the direction perpendicular to the strip. This force is oriented in the direction opposite to the electric field (see fig.2). The other force is created by the charges accumulated on the edges of the strip. They create an electric field E and the force is given by :

(2) $$F_{elec.} = nqE = nqV_H/\lambda$$

where n is the density of charge carriers, q the electric charge of them, and V_H is the Hall voltage. If we equal the two forces, we get the Hall resistance and the Hall resistivity :

(3) Hall resistance $R_H = V_H/I = B/(n\delta qc)$

Hall resistivity $\rho_H = E/j = B/(nqc)$

We see that a high resistance is obtained when the strip is very thin, and the carrier density is low enough. Moreover the sign of the Hall voltage is determined by the sign of the charges carrying the current. They are negative for the gold or the copper, but positive for the iron. This was noted by Hall in 1880. Another consequence is that the speed v of the carriers is obtained from the remark that $j = nqv$ which implies $v = cE/B$ (L. Boltzmann 1880 [21]). The only characteristic of the sample entering in this formula is the charge carrier density. This explain why this effect is so universally used.

Let us also remark that in two dimensions, namely in the limit where δ tends to zero, we do not define the resistivity nor the current density in the same way in order to take into account the divergence when $\delta \to 0$. In particular one immediately sees that the two dimensional resistivity is also measured in Ohm whereas the three dimensional one is measured in Ohm x m . Therefore a two dimensional measurement of the Hall resistivity will give a measurement of the Ohm standard without any reference to any unit of length, an advantage only if the accuracy of the Hall measurement is high enough. This was accomplished a century after Hall, with the experiment of Von Klitzing, Pepper and Dorda [62] leading to what is called nowadays the Quantum Hall effect (see below).

The next step is the L.D. Landau theory [68] developed in 1930. In a pure metal, like the copper or the gold, the charge carriers can be considered as free independent particles. They constitute thermodynamically a perfect Fermi gas, and therefore at relatively low temperature, their energy lies below the Fermi level. The motion of each particle is governed by the Schrödinger equation :

(3) $$H\psi(\mathbf{x}) = (-i\hbar\partial/\partial\mathbf{x} - q\mathbf{A}/c)^2/2m\ \psi = E\psi(\mathbf{x})$$

where $\mathbf{A}$ are the three components of the vector potential created by the magnetic field, $q = \pm e$ the carrier charge, and m their effective mass [M20]. In the two dimensional approximation, the motion in the direction perpendicular to the strip is frozen out. On the other hand if the sample is large enough, one can consider it as infinitely extended, and it is therefore identified with $\mathbb{R}^2$. The previous operator can be written as :

(4) $$H = (K_1^2 + K_2^2)/2m \qquad K_i = (-i\hbar\partial/\partial x_i - qA_i/c) \qquad i=1,2,$$

We remark that the Y's satisfy the following canonical commutation relations :

(5) $$[K_1, K_2] = i\hbar qB/c\ \mathbf{1}$$

Therefore H appears as the hamiltonian for a harmonic oscillator, and the energy spectrum is now :

(6) $$E_n = \hbar\omega_c(n+1/2) \qquad n = 0,1,2,\ldots \qquad \omega_c = |q|B/mc$$

ω_c is called the cyclotronic frequency and corresponds to the rotation frequency of the classical motion of a particle in the magnetic field B. When B–>0 the spectrum becomes continuous.

The current is now given by the vector valued operator :

(7) $$\mathbf{j} = q\,\mathbf{v} = (iq/\hbar)[H, \mathbf{x}] = q\,\mathbf{K}/m$$

Let us assume now that at time zero, an electric field $\mathbf{E}$ is turned on. Since we consider the infinite volume limit, the two directions in the plane of the sample are equivalent. Let us choose the 1-axis parallel to $\mathbf{E}$, and let ε be the modulus of E. The evolution of $\mathbf{j}$ after t=0, is now governed by the modified hamiltonian $H(\varepsilon) = H + q\varepsilon\, x_1$ namely :

(8) $$\mathbf{j}(t) = e^{iH(\varepsilon)t/\hbar}\ \mathbf{j}\ e^{-iH(\varepsilon)t/\hbar}$$

Using the Fermi-Dirac statistics, its thermodynamical average at inverse temperature $\beta=(kT)^{-1}$ is then given by :

(9) $$\langle \mathbf{j}(t)\rangle_\beta = \lim_{\Lambda\to\mathbb{R}^2} |\Lambda|^{-1}\ \mathrm{Tr}(\chi_\Lambda(\mathbf{1} + e^{\beta(H-E_F)})^{-1}\mathbf{j}(t))$$

where χ_Λ is the indicator function of the finite box Λ.

We remark that at zero temperature, each Landau level corresponds to a density of states equal to the quantity B/Φ, where $\Phi = hc/e$ is a quantum of magnetic flux. This can be seen by replacing $\mathbf{j}$ by $\mathbf{1}$ and the Fermi distribution by the eigenprojection on the given level in the right hand side of (9). If $(n\delta)$ is the charge carriers density per unit area, we see that the number of level which can be filled is $\nu = (n\delta)\Phi/B$, a quantity called the **band filling**.

It turns out that the right hand side of (9) can be computed exactly: it is made of a time periodic function of t at the cyclotronic frequency. The constant part is parallel to the 2-axis, and its amplitude is :

$$(10) \qquad |\mathbf{j}_{av}| = \lim_{\tau\to\infty} |\tau^{-1}\int_0^\tau dt \langle \mathbf{j}(t)\rangle| = (e^2/h)\varepsilon\, \nu(\beta)$$

$$\nu(\beta) = \sum_{n\geq 0} \{\mathbf{1} + e^{\beta(\hbar\omega_c(n+1/2)-E_F)}\}^{-1}$$

The resistivity is now a 2x2 matrix such that $\mathbf{E} = \boldsymbol{\rho}\mathbf{j}$ and $\boldsymbol{\rho}^{-1} = \boldsymbol{\sigma}$ is the conductivity. From the previous calculation, it follows that both matrices are antidiagonal, which means that :

$$(11) \quad \boldsymbol{\rho} = \begin{vmatrix} 0 & \rho \\ \rho & 0 \end{vmatrix} \qquad \boldsymbol{\sigma} = \begin{vmatrix} \sigma_{xx}=0 & \sigma_{xy}=\sigma \\ \sigma_{xy}=\sigma & \sigma_{yy}=0 \end{vmatrix} \qquad \sigma = e^2/h.\nu(\beta)$$

In particular, this free system is at the same time a perfect conductor and a perfect insulator in the electric field direction ! In absence of a magnetic field the situation would be just the opposite. The diagonal coefficients of $\boldsymbol{\rho}$ which in this case vanish, are called **magnetoresistivity**.

Actually a real sample is never strictly two dimensional. The motion in the magnetic field direction is governed by the kinetic part $p_3^2/2m$ with a zero boundary condition on the upper and the lower sides of the strip. Since this part commutes with H, it simply adds a discrete set of eigenvalues to each Landau level. They exhibit therefore a slight broadening but do not change the conclusion.

On the formulæ (10) & (11) one sees that the factor $\nu(\beta)$ admits two limiting behaviors. On the one hand at reasonably high temperature (i.e. for β small), the discrete sum in (10) can be replaced by an integral if $\beta\hbar\omega_c$ is small. However βE_F is generally quite big. In the limit where $\beta\hbar\omega_c \to 0$ and $\beta E_F \to \infty$ we get the classical result $\sigma = e^2/h\,\nu = (n\delta)ec/B$, whereas in the limit of low temperature (i.e. $\beta \to \infty$), $\nu(\beta)$ converges to a step function taking on integer values (see fig.3 below).

There are two kind of experimental devices [M2] which are used in the measurement of the Quantum Hall effect. The first one is the metal-oxyde-semiconductor field effect transistor (MOSFET) the other one is a

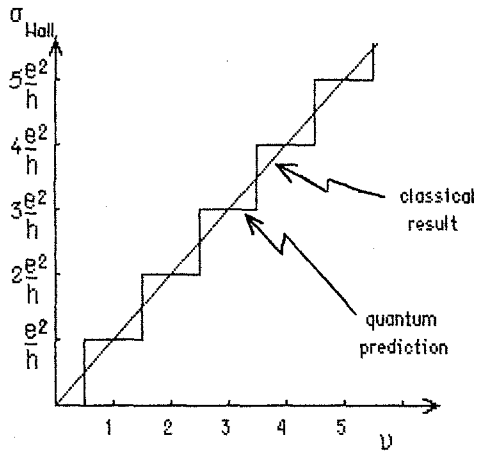

Fig. 3 : The classical Hall conductivity (high temperature) and the quantum one (at zero temperature) for a two dimensional free Fermi gas as a function of the band filling $\nu = (n\delta)hc/eB = (n\delta)\Phi/B$. Experimentally ν can be varied either by changing the magnetic field B or by changing the charge carriers density n.

heterojunction (e.g. Al_xGa_{1-x}-As-AsGa or InP-$In_xGa_{1-x}As$). In these devices one creates a thin layer, called an inversion layer, at the interface between two components, where electrons are confined and have a two dimensional motion. In a heterojunction, the principle is a little bit different [M2,97] but leads also to the existence of a thin layer of electrons with a low density. However they are practically better to use nowadays for the interface is free of impurities, and the mobility of the electrons in the inversion layer is usually much higher. Thus quantum effects are likely to appear at higher temperature.

The previous theory of the free Fermi gas must be improved in practice to take into account the influence of the crystalline structure of the substrate on the electrons. In particular, the effective potential will give rise to a broadening of the Landau levels. There is no reason that the previous result, namely the quantization of the Hall conductivity at low temperature may survive. On the other hand the direct conductivity σ_{xx} does not vanish usually, and exhibits some oscillations when varying the magnetic field or the charge density carrier : this is the de Haas-Shubnikov effect [M20]. However numerical calculations, based upon approximate theories, taking into account the disorder, and performed during the seventies [M2,2] predicted that such a quantization may survive. The question was to know with what accuracy.

In 1975, the japanese group of S. Kawaji [M2] had already observed some deviation from the classical law (11) at low temperature. The samples used at this time improved rapidly afterward, and in 1980 two groups S. Kawaji and K. Wakabayashi [59,60] in Japan and K. von Klitzing, G. Dorda and M. Pepper [62] from the Max-Planck Institut at Grenoble observed that the Hall resistance admits some plateaux when varying the band filling. The von Klitzing group also observed that these plateaux corresponded to integer multiples of e^2/h for the Hall conductivity, with an accuracy better than 10^{-5} (see fig. 5 below)! Since this time the experiment has been performed by several groups, and the accuracy is better than 10^{-7} ! Since e^2/h is a physical

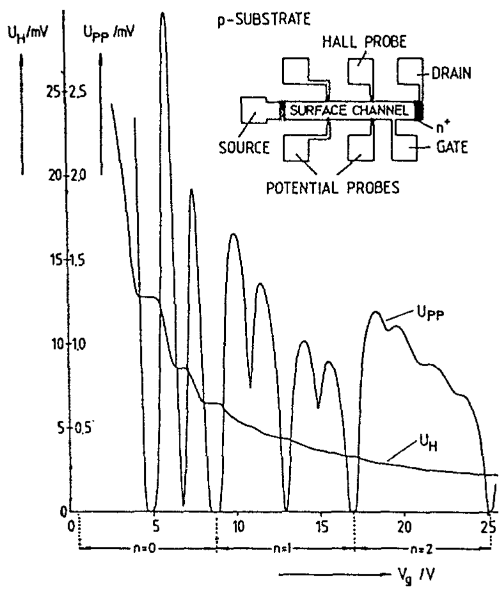

Fig. 4 : Recording of the Hall voltage, U_H , and the voltage drop between the potential probes, U_{pp} , as a function of the gate voltage at T=1.5 K. The magnetic field is 18T. The oscillation up to the Landau level n=2 is shown. The Hall voltage and U_{pp} are proportional to ρ_{xy} and ρ_{xx} respectively ($\rho=\sigma^{-1}$ see eq. 11). The inset shows a top view of the device with a length of L=400µm , a width of W=50µm, and a distance between the potential probes of L_{pp}=130µm. (Taken from ref. [62])

constant related to the fine structure constant, the Hall effect provides an independent way of checking the validity of the quantum electrodynamics. As explained already, it also provides a measurement of a universal unit of resistance ($R_H = e^2/h \approx 6453{,}2$ ohms) and a new Ohm standard.

The main surprise was not the Hall effect itself but the extremely high accuracy of the result. The idea was that it may have a very deep and universal origin. The first explanation was given by Laughlin [69] who related this quantization phenomena to the gauge invariance of the hamiltonian describing the electron. More precisely, he considered a sample having the form of a ring (see fig. 5 below) of perimeter L and width δ. He assumed that a magnetic field of constant modulus was crossing the loop perpendicularly to it. The Hall current is then given by the adiabatic derivative of the total electronic energy U of the system with respect to the magnetic flux through

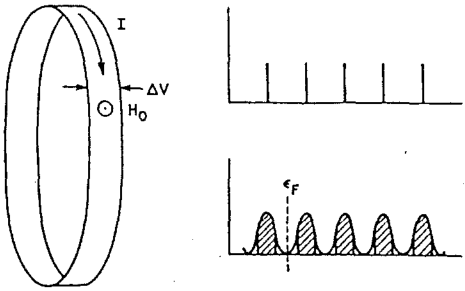

Fig.5 : Left : diagram of metallic loop. Right : density of states without (top) and with (bottom) disorder. Region of delocalized states are shaded. The dashed line indicates the Fermi level (Taken from ref. [69])

loop. This may be obtained through a "Gedanken" experiment, by adding a flux φ in the middle of the loop, and increasing it slowly in such a way that the system be constantly in equilibrium. If the additional magnetic field $\delta\mathbf{B}$ created by the flux φ, vanishes on the loop, we can describe it with a uniform vector potential $\delta A = \varphi/L$ pointing around the loop. Let x be the coordinate around the loop and y the coordinate transverse to it. We also assume that a transverse electric field $\mathbf{E}$ exists due to the Hall effect. The one-electron hamiltonian is given in the Landau gauge by :

$$(12) \qquad H(\delta A) = [(p_x - q\delta A/c + qBy/c)^2 + p_y^{\,2}]/2m^* + W(x,y) + e\mathbf{E}y$$

with periodic boundary condition with respect to x and Dirichlet boundary conditions at y=0 and y=δ. We can change this operator through a gauge transformation multiplying the wave functions by the phase factor $e^{i2\pi\alpha x/L}$ where $\alpha = \varphi/\Phi$ is the ratio between the additional flux and the quantum of flux. If α is not an integer, the new wave functions have a discontinuity at x=0. This is a unitary transformation which suppresses the δA dependence in the first term of (12) but modify the domain of H(δA). The advantage of this representation is that in this frame, H(δA) becomes a period one function of the parameter α.

If we now consider the situation where the potential W can be neglected, the one-electron wave function has the form :

$$(13) \qquad \psi(x,y) = e^{i2\pi m x/L}\; h(y-a)\; e^{-((y-a)/r)^2}$$

where h is some Hermite polynomial properly normalized. The corresponding energy is linear in a. Changing α from zero to one, namely changing the flux through the loop by one quantum unit, has the effect of changing a into $a - \Delta\delta A/B$. Since this transformation maps the system back to itself, the energy increase due to it results from a net transfer of n electrons from one edge to the other [69]. The energy increase is then given by neV where V is the

Hall voltage between the edges. On the other hand the current is given by $I = c\Delta U/\Delta\varphi = ne^2V/h$. Hence the Hall conductance (in two dimension it equals the conductivity) is an integer multiple of e^2/h.

If the system is now dirty we must assume that **the Fermi level lies in a gap of extended states**. In this case the Hamiltonian is still periodic in α and the same argument holds even though now we have no excitation of quasiparticle across the gap, and the charge is transferred only through the extended states with energy close to the Landau level as in (13). The number of electrons transferred in this way may be different but it is still an integer.

Clearly the argument must be supplemented by some more rigorous study. More recently Avron and Seiler [10] produced an argument, using [9,99], which made the Laughlin argument rigorous under the condition that the Fermi level lies in a true gap. Again in this latter work, the topology of the sample seems essential. We should be able to avoid such a constraint which does not fit with the experiment.

Several works were useful to investigate each part of the argument. In his early work, using weak-scattering calculation, Ando & Aoki [3] showed that the presence of an isolated impurity does not affect the Hall current. A similar result was obtained by Prange [83] for a δ-functions impurity, to the leading order in the drift velocity $c\mathbf{E}/B$. Later on Thouless [98], showed indeed that such an invariance holds at least for weak disorder, when a true gap occurs between Landau levels.

An important step was performed by Den Nijs, Kohmoto, Nightingale and Thouless [99] (N_2KT) who recognized that the Hall conductance for a perfect crystal is given by a homotopy invariant quantity over the two dimensional torus given by the Brillouin zone when the Fermi level lies in a gap. Avron, Seiler and Simon [9] showed that it is actually the Chern character of some fibre bundle over this torus. This torus sits in the momentum space rather than in the real space, and is independent of the topology of the sample. The very high stability of the result under perturbation is explained by the homotopy invariance of the Chern character. However, they required the magnetic field to satisfy a rationality condition, namely that the flux through one unit cell be a rational multiple of the quantum of flux. Up to now, no rigorous proof is available yet, when the Fermi level lies in a gap of extended states, and the topology of the sample is arbitrary.

8)- The quantum Hall effect and Connes's cohomology :

In this section, we intend to describe a mathematical framework using C*-algebras in order to give a local proof of the quantum Hall effect, independent of the topology of the sample. We follow the idea of the N_2KT paper to express the Hall conductivity in term of a non commutative Chern character. The main approximations are the following :

- we consider only the one electron theory.
- we work with a 2D disordered sample of infinite size.

In this approach however, we shall miss one point, essential for a physicist, namely, that the result will be valid only if the Fermi level belongs to a gap of the spectrum of the one body hamiltonian. We shall indicate what should happen if the Fermi level belongs only to a gap of extended states.

The Schrödinger operator for a charge carrier in the sample submitted to a uniform magnetic field perpendicular to it is given by :

$$(1) \qquad H_\omega = 1/2m[(p_1-eA_1/c)^2 + (p_2-eA_2/c)^2)] + V_\omega(x_1,x_2) = H_0 + V_\omega$$

where m is the effective mass of the charge carrier, e its electric charge, and ω represents the effect of the impurities on the potential. As previously, ω will be taken in a compact metrisable space Ω with :

$$(2) \qquad V_\omega(x_1,x_2) = v(T_{-\mathbf{x}}\omega) \qquad v \in C(\Omega) \qquad \mathbf{x} = (x_1,x_2) \in \mathbf{R}^2$$

Let us introduce the vector valued operator $\mathbf{K} = \mathbf{p} - e\mathbf{A}/c$ and for $\boldsymbol{\xi} \in \mathbf{R}^2$ and f a function in the Schwartz space $S(\mathbf{R}^2)$, we set :

$$(3) \qquad W(\xi) = \exp(i\xi\mathbf{K}/\hbar) \qquad W(f) = \int_{\mathbf{R}^2} d^2\xi\ f(\xi)\, W(\xi)$$

These "Weyl operators" fulfill the following commutation relations :

$$(4) \qquad W(\xi)\, W(\xi') = W(\xi+\xi')\, e^{i\pi\alpha\xi\wedge\xi'} \qquad \xi\wedge\xi' = (\xi_1\xi'_2 - \xi'_1\xi_2)$$

$$\alpha = B/\Phi = eB/hc$$

It is a well known fact that the C*-algebra generated by the W(f)'s is isomorphic to the algebra of compact operators $\mathcal{K}$ on a separable Hilbert space. Moreover the resolvent of the free part of the hamiltonian belongs to this algebra. We shall denote by $\mathcal{A}$ the C*-algebra generated by operators of the form $V_\omega W(f)$ with V as in (2) and f in the Schwartz space.

This algebra can be abstractly reconstructed as follows [see M23] : we consider the space of continuous functions with compact support on $\Omega \times \mathbf{R}^2$ endowed with the following structure of *algebra :

$$(5)\qquad ab(\omega,\xi) = \int_{\mathbf{R}^2} d^2\xi' \; a(\omega,\xi')b(T_{-\xi'}\omega,\xi-\xi')\, e^{i\pi\alpha\xi\wedge\xi'}$$

$$a^*(\omega,\xi) = a(T_{-\xi}\omega,-\xi)^*$$

We built $\mathcal{A}$ as in the section 2 (equ.5) with the representations π_ω acting on $L^2(\mathbf{R}^2)$:

$$(6)\qquad \pi_\omega(a)\psi(\mathbf{x}) = \int_{\mathbf{R}^2} d^2\xi \; a(T_{-\mathbf{x}}\omega,\xi-\mathbf{x})\, e^{i\pi\alpha\xi\wedge\mathbf{x}}\, \psi(\xi)$$

One can check that the resolvent $R_\omega(z)=(z-H_\omega)^{-1}$ of the hamiltonian H_ω has the form $\pi_\omega(r(z))$ for some $r(z)$ in $\mathcal{A}$.

If now μ is a T-invariant ergodic probability measure on Ω, one can define a trace τ_μ on $\mathcal{A}$ as before by the formula (3) of the section 4. In addition we have a differential structure [28,32,33] through the data of the following two derivations of $\mathcal{A}$:

$$(7)\qquad \delta_j a(\omega,\xi) = 2i\pi\xi_j a(\omega,\xi) \qquad\qquad j=1,2$$

or

$$\pi_\omega(\delta_j a) = 2i\pi\,[\,\pi_\omega(a), Q_j]$$

where the Q_j's are the operators of multiplication by x_j in $L^2(\mathbf{R}^2)$.

The main remark of Thouless et al. in [99] consists in writing the Hall conductivity by mean of the Kubo formula, in a way which identifies it with a Chern character. In our notation this gives the following :

<u>Proposition 11</u> : The expression of the Hall conductivity according to the Kubo formula at zero temperature is given by :

$$(8)\qquad \sigma_H = e^2/h\,(1/2i\pi)\,\tau_\mu(P_F[\delta_1 P_F,\delta_2 P_F])$$

where $\pi_\omega(P_F)$ is the eigenprojection of H_ω on energies smaller than or equal to the Fermi energy E_F. This formula is valid provided the Fermi energy lies in a gap of H_ω (μ-almost surely). ◇

Before using this result, let us comment about the way this formula is derived.

1)-The proof of it is just a matter of calculation once the Kubo formula is accepted. To compute the Kubo formula however, one should use the following steps : (i) as in the section 7 (eq. 7-10) one must compute the current at time t after turning on an electric field ε in the x_1 direction ; (ii) at zero temperature, the Fermi Dirac function reduces to the projection P_F ; (iii) one computes the time average of the statistical average of the current, and (iv) we compute the first order term as the electric field ε goes to zero. Actually, in practice one usually exchanges the time averaging and the limit $\varepsilon \to 0$. The control of this exchange has been performed in this particular case by R. Seiler [92], for a finite size sample. Since there is no thermal dissipation in this problem, the difficulties in doing this seems purely technical. One does not know how to extend the Seiler proof in our framework case.

2)-In the course of the calculation, one uses the fact that the Green function of H_ω, decreases exponentially fast at infinity when the Fermi energy lies in a gap. Following the arguments given by Prange [83], Thouless [98], and Halperin [50], it is likely that the calculation extends to the case for which the Fermi level lies in the pure point spectrum, for in this case, since the states of energy close to E_F are localized, the Green function still decreases fast enough at infinity to insure the convergence of the integrals. However, as shown by Fröhlich and Spencer [44], one must be careful with "almost sure" properties.

3)-It has been proved rigorously recently [20] that in 2D, in the framework of a tight-binding approximation, all states are localized at high disorder, and the spectrum is pure point. At lower disorder, only the states corresponding to energies in the band edges are localized.

Once we are ready to admit the previous formula, one remarks with A. Connes that the expression :

$$(9) \qquad Ch(P) = (1/2i\pi)\, \tau_\mu(P[\delta_1 P, \delta_2 P])$$

where P is a projection in $\mathcal{A}$, satisfies the following properties :

$$(10) \quad \begin{array}{llll} (i) & \text{if } P \approx Q & \text{then} & Ch(P) = Ch(Q) \\ (ii) & \text{if } PQ = QP = 0 & \text{then} & Ch(P \oplus Q) = Ch(P) + Ch(Q) \end{array}$$

This shows that Ch actually defines a mapping on the K-group of $\mathcal{A}$, and that it is a homomorphism. It is therefore sufficient to know a set of generators of the K-group in order to know what are the possible values of Ch(P). Thank to the proposition 11, the Quantum Hall effect will be established in this set up once we are able to answer yes to the following question :

Question 5 : Let $(\Omega, T, \mathbf{R}^2, \mu)$ be a dynamical system as before, and let $\mathcal{A}(\Omega \times_T \mathbf{R}^2, \alpha)$ be the corresponding algebra constructed according to eqs. (5-6). Then is the Chern character Ch(P) of any projection in this algebra an integer ?

The answer to this question is yes in a certain number of cases. First of all, if Ω is reduced to a point the algebra is nothing but the algebra of compact operators and the answer is trivially yes. More involved is the case for which V is periodic. In this case, Ω is a two dimensional torus $\mathbf{T}^2$ and the algebra is isomorphic to the stabilized the rotation algebra $\mathcal{A}_\alpha$. The answer in this case is still yes thanks to the Pimsner & Voiculescu [82,86], Rieffel [] who computed the two generators of the K-group, and to an explicit calculation of A. Connes [30], who computed their Chern character. Note that in this previous case, the spectrum is likely to be a Cantor set [18,23,53], and the gaps are not necessarily associated with Landau levels. At last, the answer is still yes at small disorder, for in this case, H_ω is close to an operator (in the norm sense) for which the Chern character can be explicitly computed. Using the homotopy invariance of Ch(P), the value of Ch(P_F) is still an integer as far as E_F stay in a gap while turning on the disorder.

Let us note at last the two homotopy invariance results :

<u>Proposition 12</u> : (i) Let us assume that the potential V in (1) depends continuously (in the norm sense) upon a parameter λ. Then the Chern character Ch($P_F(\lambda)$) is independent of λ as far as the Fermi level stay in a gap.
(ii) The Chern character Ch(P_F) is continuous in α (i.e. of the magnetic field) as far as the Fermi level stay in a gap. ◇

The first result is just the usual homotopy invariance of the class of K-theory of a projection. The second one can be shown by considering the universal algebra obtained as the union over α of the previous ones [41]. Then the trace τ_μ can be shown to be a pointwise continuous function of α.

APPENDIX

Shubin's formula for a discrete action

Let G be a discrete, countable amenable group. Let Ω be a compact space on which G acts by a family of homeomorphisms such that the map $(\omega,x) \to x^{-1}\omega$ be continuous. Let μ be a G-invariant ergodic probability measure on Ω and as in the §4, let τ_μ be the corresponding trace on $C^*(\Omega,G)$. Let H be a bounded self adjoint operator on $l^2(G)$ and ω_0 in Ω, h in the C*algebra of the dynamical system (Ω,G) such that :

(1) $$H = \pi_{\omega_0}(h)$$

For Λ a finite subset of G let χ_Λ be the projection onto $l^2(\Lambda)$ in $l^2(G)$, and let H_Λ be the restriction of H on Λ namely

(2) $$H_{\omega,\Lambda} = \chi_\Lambda \, \pi_\omega(h) \, \chi_\Lambda$$

Since Λ is finite, H_Λ is finite dimensional, and its dimension is $|\Lambda|$, the cardinality of Λ.

The density of states was defined in the §4 eq.2, and gives rise to a probability measure defined by :

(3) $$\int d\mathcal{N}_\omega(E)\, f(E) = \lim_{n\to\infty} |\Lambda_n|^{-1} \mathrm{Tr}\,(f(H_{\omega,\Lambda_n}))$$

where Λ_n is a Følner sequence in G. In the sequel we will drop the index n. The first result we need is :

<u>Lemma 1</u> : for μ almost every ω, and all $f \in C(\Omega)$, the limit (3) exists and is equal to :

(4) $$\int d\mathcal{N}_\omega(E)\, f(E) = \tau_\mu\,(f(h))$$

◇

<u>Proof</u> : For any finite Λ the right hand side of (3) defines a probability measure (Riesz's theorem) on the convex hull of the spectrum of h, for if we denote it by $\mathcal{N}_\Lambda(f)$, we get

(i) $\mathcal{N}_\Lambda(f)$ is linear in f

(ii) $\mathcal{N}_\Lambda(1) = 1$ and $\mathcal{N}_\Lambda(f) \geq 0$ if $f \geq 0$

Consequently we have

(iii) $|\mathbf{N}_\Lambda(f)| \leq \| f \|$ for all Λ

On the other hand from the definition of the trace given in §4 eq.5 we have

$$(5) \qquad \tau_\mu(f(h)) = \lim_{|\Lambda| \to \infty} |\Lambda|^{-1} \mathrm{Tr}\,(f(H_\omega)\chi_\Lambda) \text{ for } \mu \text{ almost all } \omega$$

In much the same way, for each Λ, the right hand side of (5) defines another probability measure on the convex hull of the spectrum of h which we denote by $\tau_\Lambda(f)$. We know from the Birkhoff theorem [M13], that the limit exists μ almost surely. The result will be achieved if we prove that the difference $\mathbf{N}_\Lambda(f) - \tau_\Lambda(f)$ converges to zero as $|\Lambda|$ goes to infinity.

Using the Stone-Weïerstrass theorem it is sufficient to prove the result when f is a monomial. Let k be a positive integer. Then dropping the index ω (for k=1 we get zero) :

$$(6) \qquad |\Lambda| \{\mathbf{N}_\Lambda(X^k) - \tau_\Lambda(X^k)\} = \mathrm{Tr}\{\chi_\Lambda(H^k - (\chi_\Lambda H \chi_\Lambda)^k)\} =$$

$$= \sum_{1 \leq j \leq k-1} \mathrm{Tr}\{\chi_\Lambda H^j (1-\chi_\Lambda)(H\chi_\Lambda)^{k-j}\} \leq (k-1) \| H \|^{k-2} \mathrm{Tr}\{\chi_\Lambda H(1-\chi_\Lambda)H\chi_\Lambda\}$$

Since $H = \pi(h)$ with $h \in C^*(\Omega,G)$, for any $\varepsilon > 0$, there is $h(\varepsilon)$ in $C_c(\Omega \times G)$ such that $\| H - H(\varepsilon) \| < \varepsilon$, if $H(\varepsilon) = \pi(h(\varepsilon))$. Replacing H by $H(\varepsilon)$ in the right hand side produces an error $\delta = (k-1) \| H \|^k \varepsilon |\Lambda|$. On the other hand we get :

$$(7) \qquad \mathrm{Tr}\{\chi_\Lambda H(\varepsilon)(1-\chi_\Lambda)H(\varepsilon)\chi_\Lambda\} = \sum_{x \in \Lambda} \sum_{y \notin \Lambda} |h(\varepsilon; x^{-1}.\omega, x^{-1}y)|^2$$

Now since G is discrete and $h(\varepsilon)$ has a compact support, there is Γ finite in G such that $x^{-1}y \notin \Gamma$ implies $h(\varepsilon; x^{-1}.\omega, x^{-1}y) = 0$ for all $\omega \in \Omega$. Let C be the maximum of $h(\varepsilon; \omega, x)$ over $\Omega \times G$, then :

$$(8) \qquad \mathrm{Tr}\{\chi_\Lambda H(\varepsilon)(1-\chi_\Lambda)H(\varepsilon)\chi_\Lambda\} \leq C\,|\Lambda \Delta (\Lambda a)|$$

Therefore, for any $\varepsilon > 0$, there is $C(\varepsilon) > 0$, such that :

$$(9) \qquad \{\mathbf{N}_\Lambda(X^k) - \tau_\Lambda(X^k)\} \leq (k-1) \| H \|^k \varepsilon + C(\varepsilon)(k-1) \| H \|^{k-2} |\Lambda \Delta (\Lambda\Gamma)| / |\Lambda|$$

Since the group G is amenable, Γ is finite and Λ is a member of a Følner sequence, the last term of the right hand side converges to zero as $|\Lambda| \to \infty$. Since ε is arbitrary, we achieve the result.

◇

Lemma 2 : The function $E \to \tau_\mu(\chi(h \leq E))$ is increasing from 0 to 1, and is locally constant outside the spectrum of h. If E is a point of continuity of this function, then :

(10) $$\mathbf{N}_\omega(E) = \tau_\mu(\chi(h \leq E)) \qquad \text{for } \mu \text{ almost all } \omega$$

◇

Proof : The first assertion is obvious. The left hand side of (10) is defined through the weak the limit of a sequence of probability measures on the convex hull of the spectrum of h, as was proved in lemma1. The result is a consequence of a well known result on weak limits of probability measures on a a complete metric space. ◇

Remark : In general the function $\mathbf{N}(E)$ is not continuous. For if h is a projection in $C^*(\Omega,G)$, then clearly, $\mathbf{N}(E)$ is a step function with a discontinuity at E=1. Conversely, if it is discontinuous at E, then E is an eigenvalue of H of infinite multiplicity and the eigenprojection is actually in the C* algebra. The question is whether such projections exist. The answer depends upon the C*algebra. For an abelian virtual group $(\mathbf{T}^\nu, \mathbf{Z}^D)$ the answer is yes. This was proved firstly by M. Rieffel [86] on the irrational rotation algebra, and his argument extends in the general case. He even exhibited a projection given by a continuous function with compact support. However, there are examples of dynamical systems $(\Omega,\mathbf{Z})$ such that the corresponding C*algebra has no projection and is simple [29] (see also §5)

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